Data Science

Lecture 02: Probability Theory

Prof. Dr. Michael Granitzer

Where We Left Off

Lecture 01 ended on a claim: data does not speak for itself — an assumption always speaks for it, and a pattern is knowledge only once it has been checked. Today builds the language in which both are written.

Assume

Modules 1–5 — a model of chance: outcomes, probabilities, random variables, the joint

→

Observe

Module 6 — a sample, described, with its error

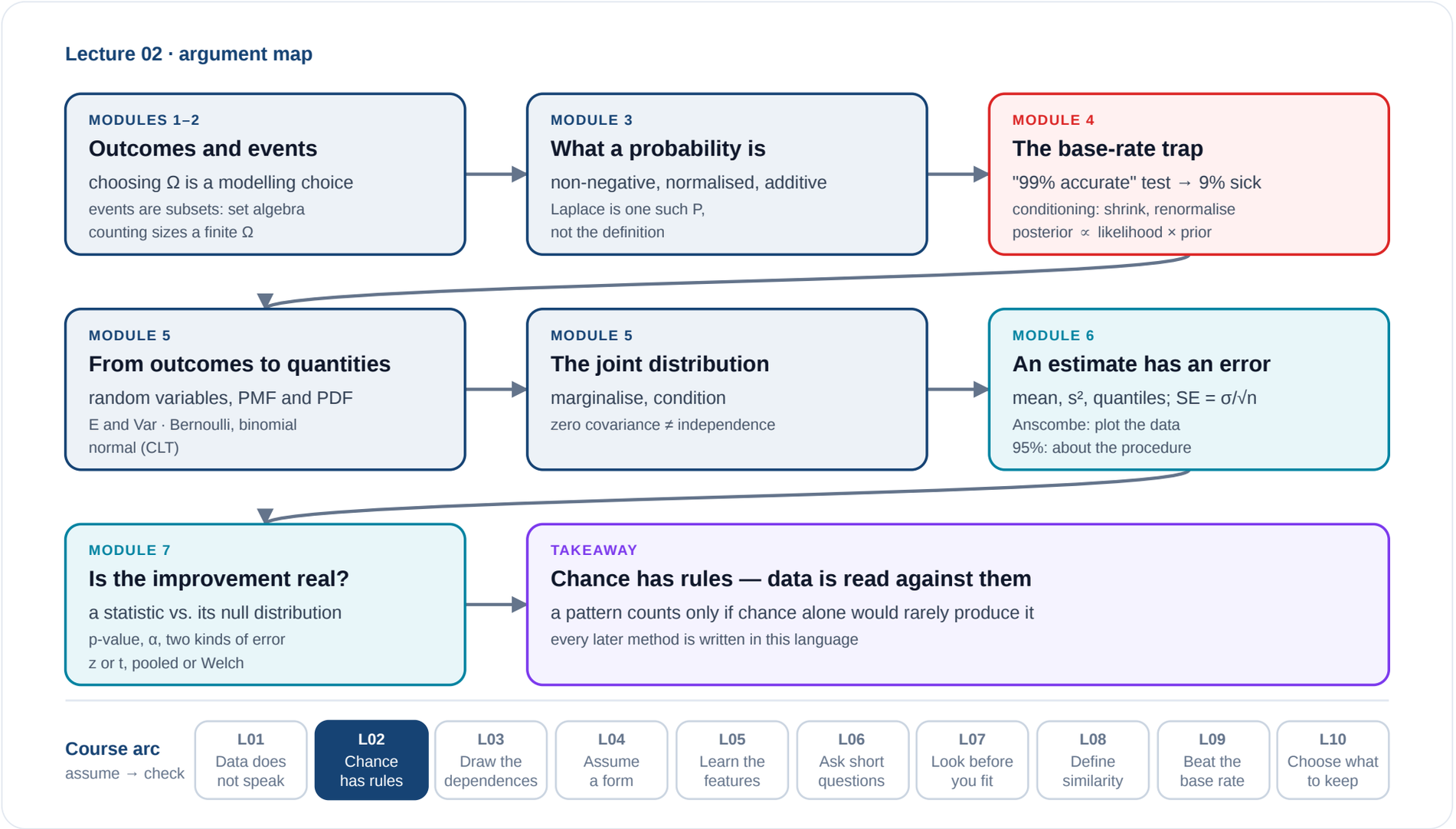
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Check

Module 7 — would chance alone produce this?

> The app users of Lecture 01 churn less. By the end of today you can say whether that difference is real — not yet *why* (that is Lecture 03).

Argument Map



Module 0

Orientation

> ***Guiding Question:*** *What does a data scientist actually need from probability theory — and why does the formal treatment come first?*

The Landscape

Probability theory
Given a model of chance, what follows?
Kolmogorov axioms, distributions, limit theorems

Mathematical statistics
Given data, which model of chance is plausible?
Estimation, hypothesis testing

Data science
Given data and a decision, what should we do?
Prediction, learning, evaluation

The arrow runs left to right: this lecture builds the vocabulary the rest of the course spends its time using.

What We Need From Probability

In this lecture

- sample spaces and events
- counting outcomes
- probability as a measure
- conditioning and independence
- Bayes' theorem
- random variables and distributions
- describing data numerically
- testing a hypothesis

Where it comes back

- naive Bayes and Bayesian decisions
- impurity measures in decision trees
- likelihoods in probabilistic models
- sampling and error estimates in evaluation
- independence assumptions in graphical models



Example: Students, Courses and Exams

Probabilistic ML — 40 students
a quarter of them fail the exam

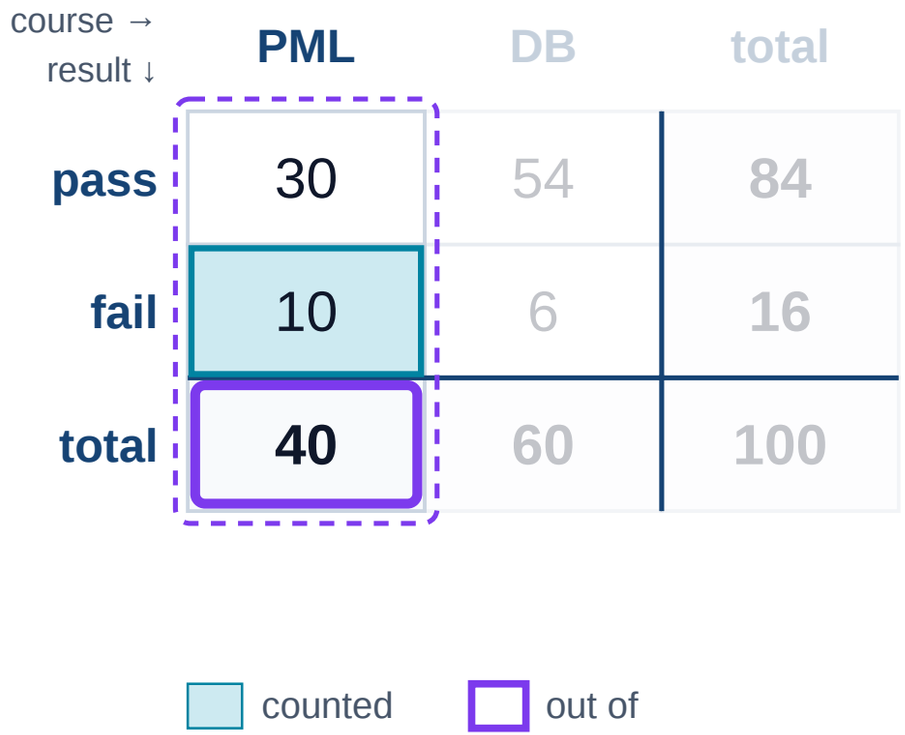
Databases — 60 students
one in ten fails the exam

Each of 100 students sat exactly one of the two exams.

- **Forward:** how many students fail overall?
- **Backward:** a student failed — how probable is it that they took PML?

Joint, Marginal, Conditional — One Table

? Of the PML students, what share failed?



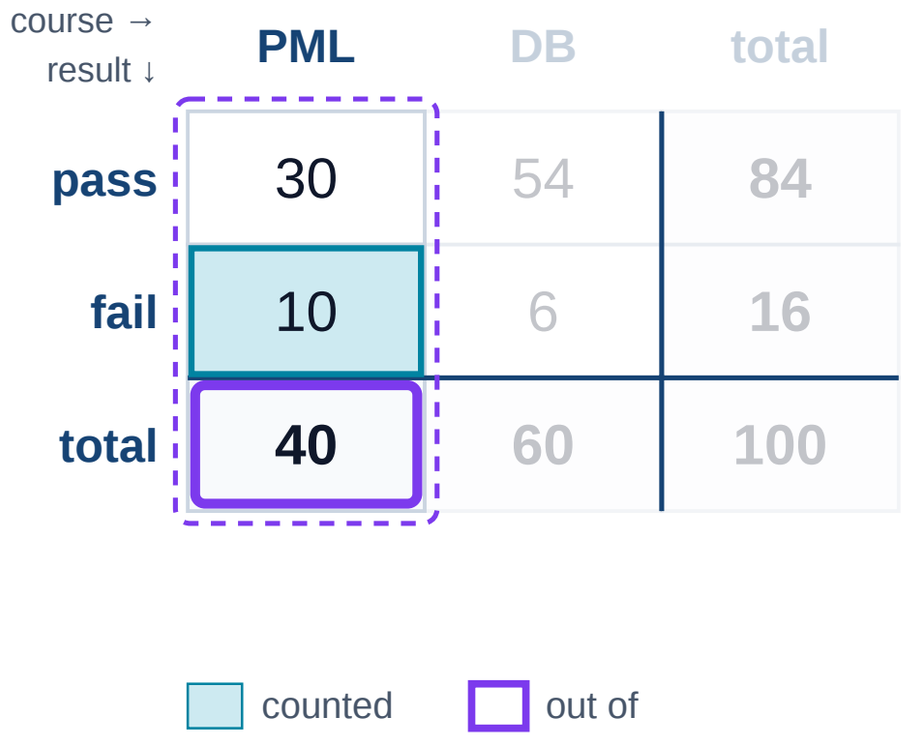
P(fail | PML)

= **10** / **40** = 0.25

conditional: a cell, out of its own column

Sum Rule and Product Rule — On the Table

? 40% took PML, and a quarter of them failed. What share failed PML?



… of those, failed: P(fail | PML) = **10** / **40**

Product rule

P(PML, fail) = P(PML) · P(fail | PML)

= 40/100 · 10/40 = 10/100

a share of a share — the 40s cancel

The Two Rules

Sum rule

$$P(X) = \sum_{Y} P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

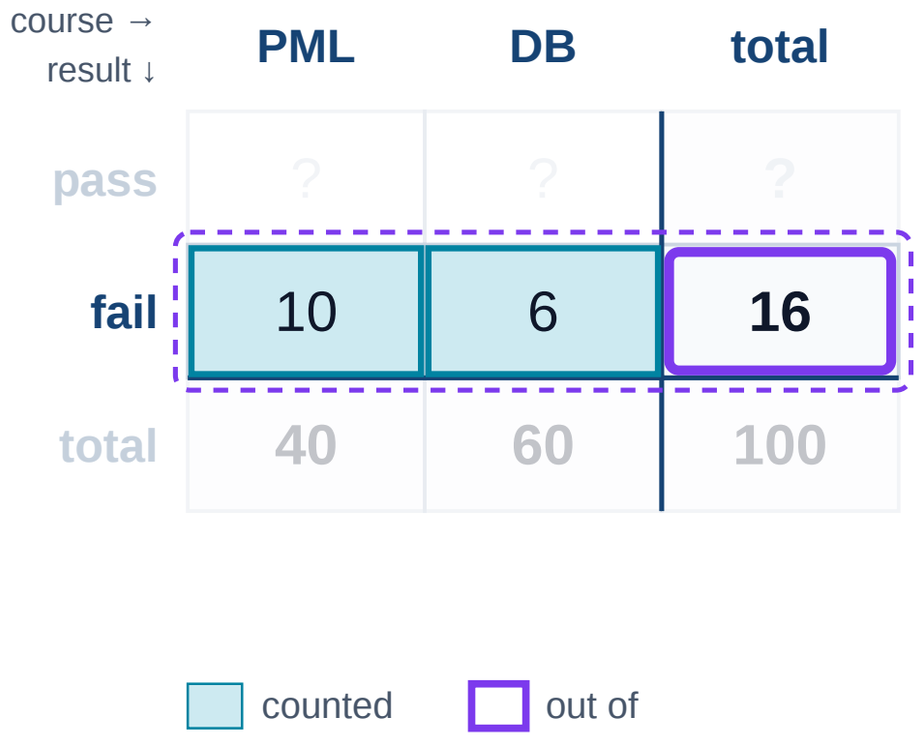
$$P(X, Y) = P(Y \mid X) P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

Bayes' Theorem — On the Table

? And which course did they most probably take?



P(PML | fail) = **10** / **16** ≈ 0.63

P(DB | fail) = **6** / **16** ≈ 0.37

PML — odds 10 : 6

same normaliser: comparing needs only the cells

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

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And if one in four DB students failed too?

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And if one in four DB students failed too?

Then $P(\text{PML} \mid \text{fail}) = 0.4$: the result says nothing about the course.

Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. What follows makes them precise and extends them beyond counting.

Module 1

Events and the Algebra of Sets

> ***Guiding Question:*** *If probability is going to be a measure, what exactly are the objects it measures — and how do they combine?*

What Is a Random Experiment?

A coin flip, tomorrow's rainfall, whether a user clicks a link. What makes each of them a *random experiment* — and what would not be one?

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> **Definition (Random experiment).** A procedure that can — at least in theory — be repeated infinitely often, characterised by:
>
> 1. **Configuration** — a precisely specified system that can be reconstructed.
> 2. **Procedure** — an instruction for executing the experiment.
> 3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

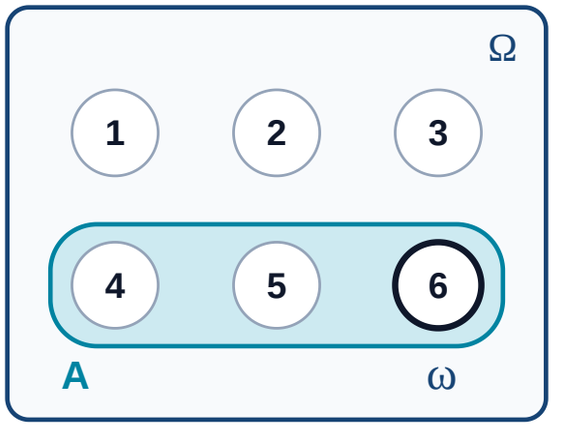
Randomness as Missing Information

> A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.
>
> Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Sample Space and Events

Definition (Sample space, event space). A set $\Omega = \{\omega_1, \ldots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$, an **outcome**.
For finite Ω, every subset $A \subseteq \Omega$ is an **event**; A **occurs** iff the observed $\omega \in A$. All events together, $\mathcal{P}(\Omega)$, form the **event space**.

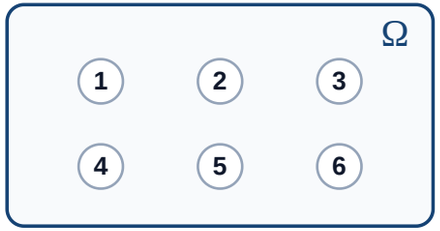


Ω	sample space: all outcomes
ω = 6	an outcome: an element
A = {4, 5, 6}	an event: a subset

ω ∈ A: the event "at least 4" occurs

An outcome is an element; an event is a subset — a *question you can ask* about the result.

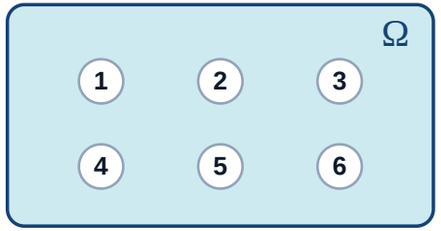
Important Event Types



$\varnothing$

impossible event

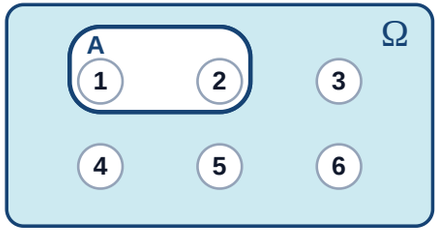
no outcome belongs to it



Ω

certain event

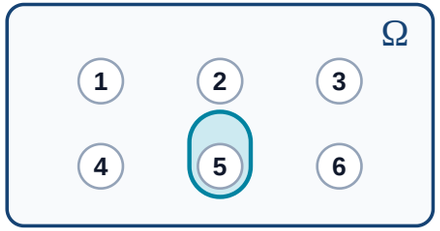
every outcome belongs to it



$\bar{A} = \Omega \setminus A$

complementary event

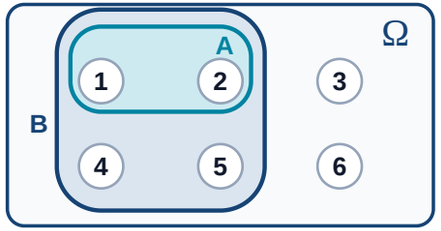
A = {1, 2}, $\bar{A}$ = {3, 4, 5, 6}



$|A| = 1$

elementary event

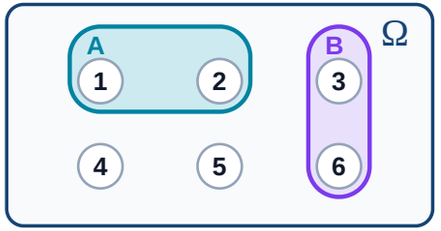
A = {5}: one outcome



$A \subseteq B$

A entails B

A = {1, 2}, B = {1, 2, 4, 5}



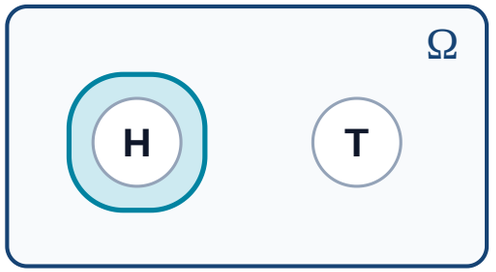
$A \cap B = \varnothing$

incompatible events

A = {1, 2}, B = {3, 6}

Examples

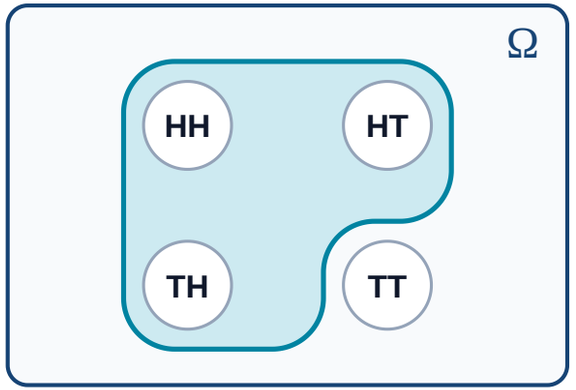
One coin flip



E = {H}: "heads"

events: ∅, {H}, {T}, {H, T}

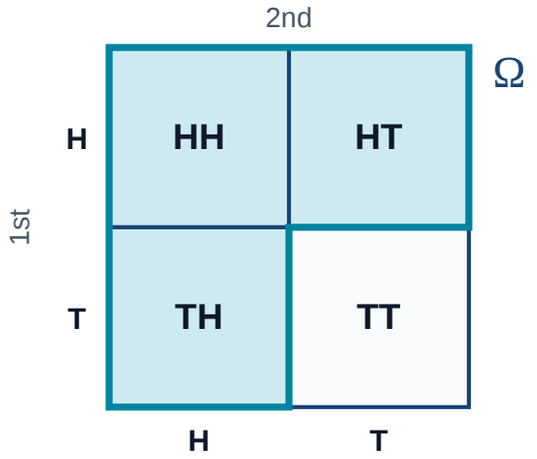
Two coin flips: the set



E = {HH, HT, TH}: "at least one head"

4 outcomes, $2^4 = 16$ events

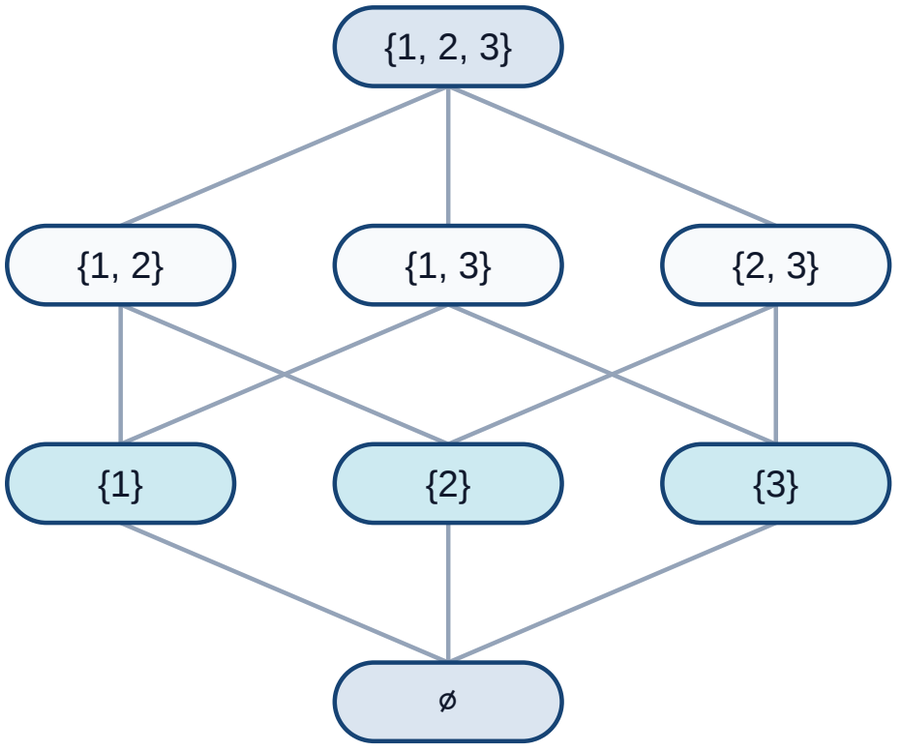
… as a grid: 1st × 2nd



same outcomes, arranged by flip

The same event, twice: as a subset of Ω, and as cells of the 1st × 2nd grid. The sample space fixes which questions can be asked.

The Event Space Grows Fast



$|\Omega| = n \rightarrow 2^n$ events

1	2
2	4
3	8
6	64
10	1,024
20	1,048,576

lines: one element added

For $|\Omega| = n$ the event space has 2^n elements — far too many to list, which is why probability is done by counting, not by enumeration.

Not Every Sample Space Is Finite

Dosage of a drug: Ω = (0, ∞)



uncountably many outcomes — probability can only live on intervals, not on points

Race with 7 teams: all orderings

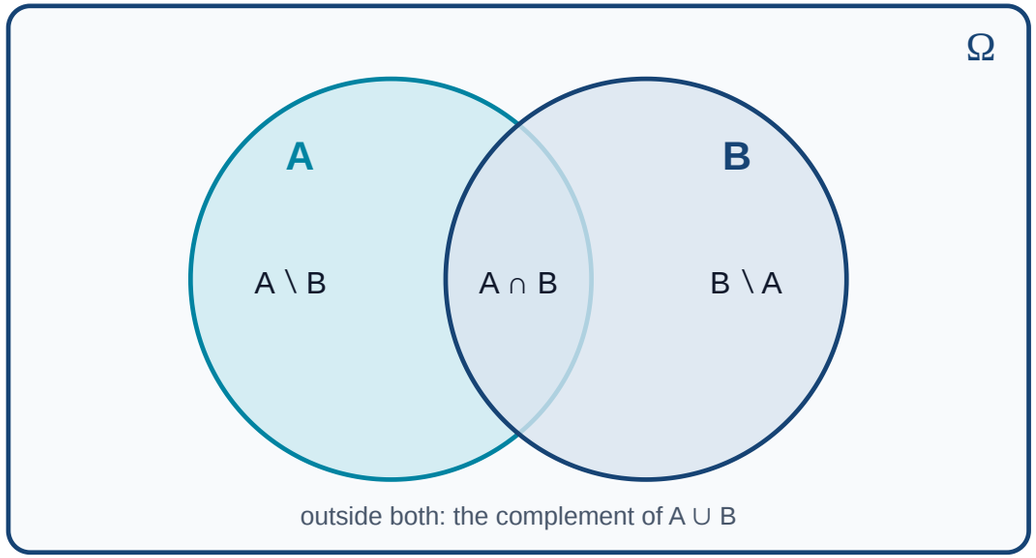


7! = 5,040 outcomes; shaded: the event "team 3 wins"

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Venn Diagrams

Venn diagrams are the graphical representation of events:

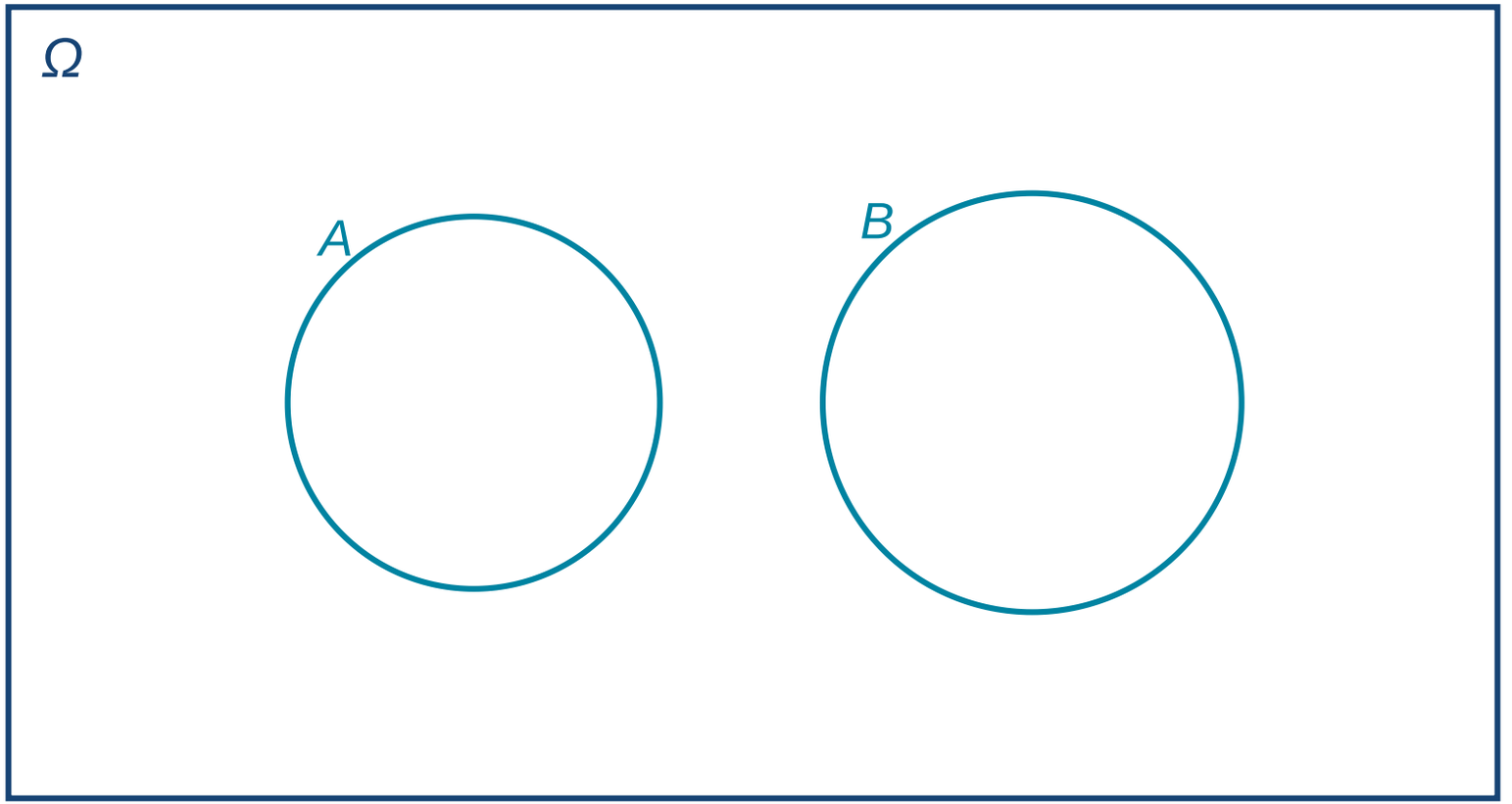


- the **rectangle** is the sample space Ω
- a **circle** is an event $A \subseteq \Omega$
- every **region** is a set — and every set operation is a region you can point at

Important Event Types

Before each step: which region will light up?

$$A \cap B = \emptyset$$



A and B are disjoint

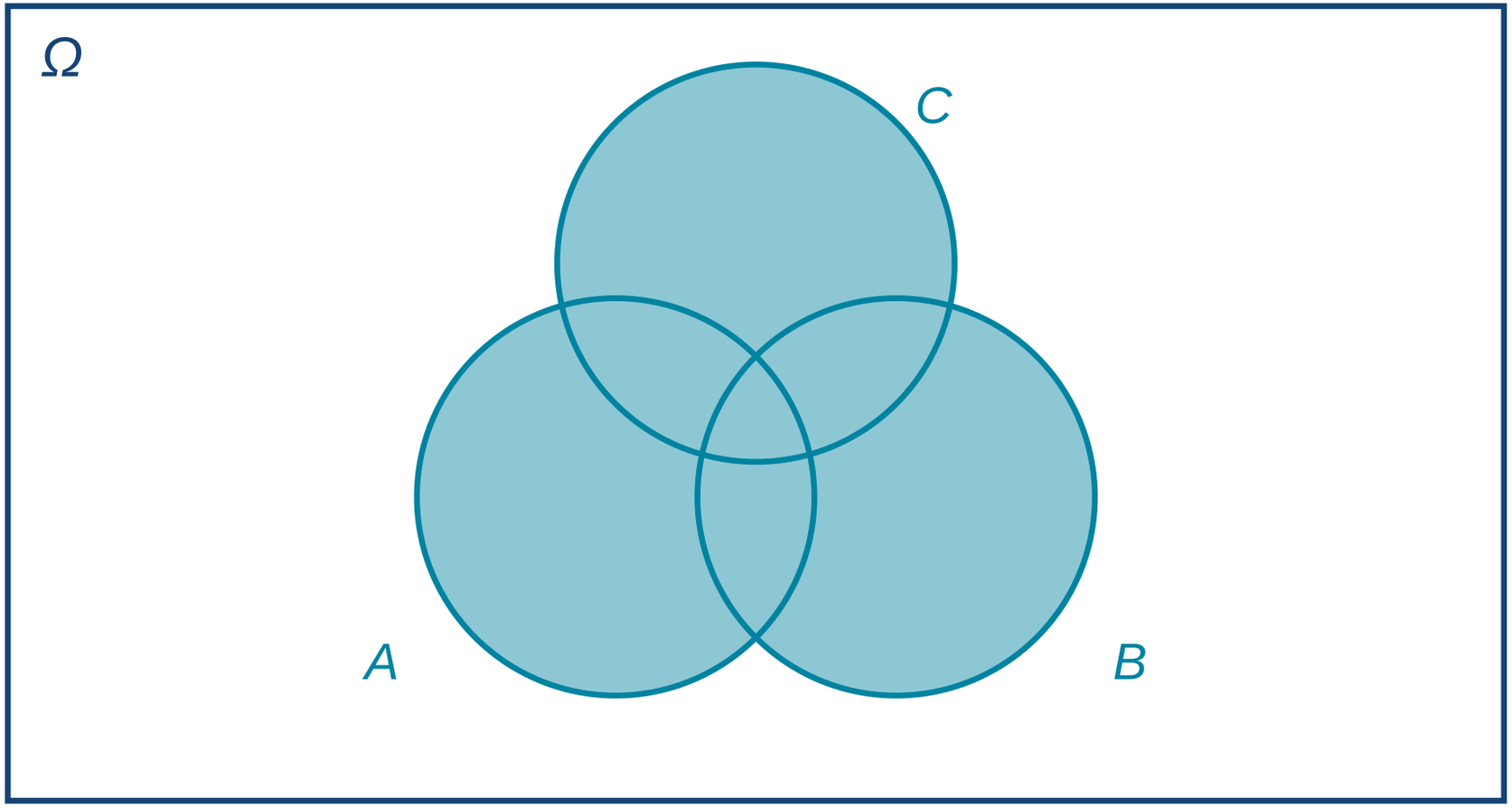
Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions

A ∪ B ∪ C



Union of three events

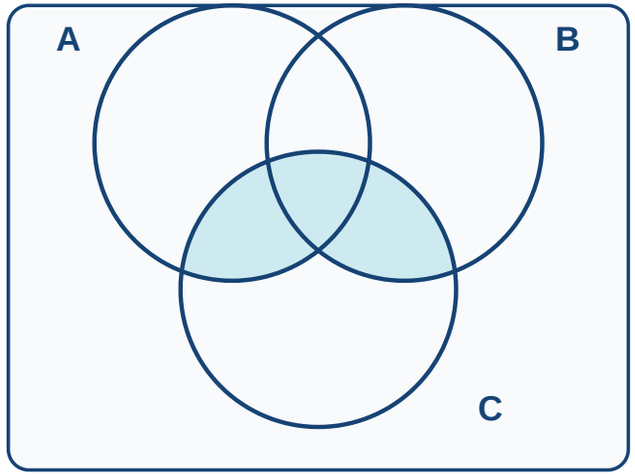
Mini-Exercise

Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

Mini-Exercise

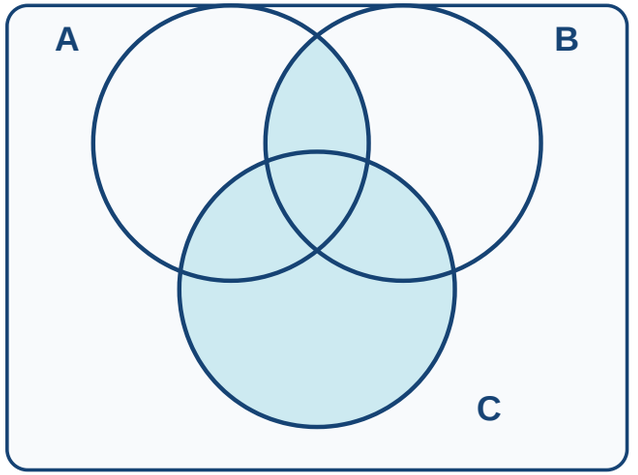
Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

(A ∪ B) ∩ C



inside C only

(A ∩ B) ∪ C



all of C, plus the lens A ∩ B

No. The first lies entirely inside C; the second is all of C plus the lens $A \cap B$. The distributive laws expand them into different expressions.

Takeaway

An event is a **subset** of the sample space, and combining statements about outcomes is set algebra: *or* is a union, *and* an intersection, *not* a complement. Choosing Ω decides which events exist at all.

Module 2

Counting

> ***Guiding Question:*** *How do we determine the size of a sample space we could never write down?*

Why Counting Comes First

If every outcome in a finite sample space is equally likely, then the probability of an event is a **ratio of counts**:

$$P(A) = \frac{|A|}{|\Omega|}$$

So before we can measure probability, we must be able to count $|A|$ and $|\Omega|$ — often without enumerating them.

Basic Principle of Counting

Generalised basic principle of counting. If an experiment consists of r stages, where stage i has n_i possible outcomes regardless of the earlier stages, then the whole experiment has

$$n_1 \cdot n_2 \cdots n_r$$

possible outcomes.

Example. How many 7-place licence plates, if the first 3 places are letters (26) and the last 4 are digits (10)?

$$26^3 \cdot 10^4 = 175{,}760{,}000$$

Three Questions Decide the Formula

Order?

Does the sequence matter?
matters → **permutation** (rankings)
does not → **combination** (committees)

Repetition?

May an element be picked twice?
without — teams of people
with — letters in a word

How many?

Select k of n elements, or arrange all n?

The Four Basic Cases

Ordered, without repetition

k-permutation of n:

$$\frac{n!}{(n-k)!}$$

From $\{a, b\}$, $k = 2$: ab, ba

Ordered, with repetition

$$n^k$$

From $\{a, b\}$, $k = 2$: aa, ab, ba, bb

Unordered, without repetition

k-combination of n:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

From $\{a, b\}$, $k = 2$: $\{a, b\}$

Unordered, with repetition

$$\binom{n+k-1}{k}$$

From $\{a, b\}$, $k = 2$: $\{a, a\}, \{a, b\}, \{b, b\}$

Permutations

Definition (k-permutation). An ordered selection of k elements from a set of n distinct elements without repetition. Counting position by position: n choices, then $n-1$, …, down to $n-k+1$, giving $\frac{n!}{(n-k)!}$.

Example. A class of 6 men and 4 women is ranked after an exam, no ties. How many possible rankings? $10! = 3{,}628{,}800$.

Combinations

Each k-combination corresponds to $k!$ different k-permutations — the same k elements in every possible order. Therefore

$$\binom{n}{k} = \frac{\text{number of } k\text{-permutations}}{k!} = \frac{n!}{k!(n-k)!}$$

Example. A committee of 3 from 20 people: $\binom{20}{3} = 1140$.

Dividing by $k!$ is the standard move for "forgetting" an ordering we counted.
Remember it as *over-count, then correct*.

Properties of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Choosing which k to include is the same as choosing which $n-k$ to leave out.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Split by a fixed element i: groups **containing** i, plus groups **not containing** i.

Binomial theorem.

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Application: How Many Subsets?

A set with n elements has

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

subsets — sum over all subset sizes.

> **Alternative intuition.** Each element is either in the subset or not: a binary choice, n times, so $2 \cdot 2 \cdots 2 = 2^n$.

This is the size of the event space $\mathcal{P}(\Omega)$ from the sample-space unit.

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

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From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Chair and deputy: order matters, $5 \cdot 4 = 20$. Committee: each pair was counted twice (as ab and ba), so $20/2! = 10 = \binom{5}{2}$.

Takeaway

> Counting reduces to three questions — **order? repetition? how many?** — and one habit: when an ordering was counted that should not have been, divide it out.

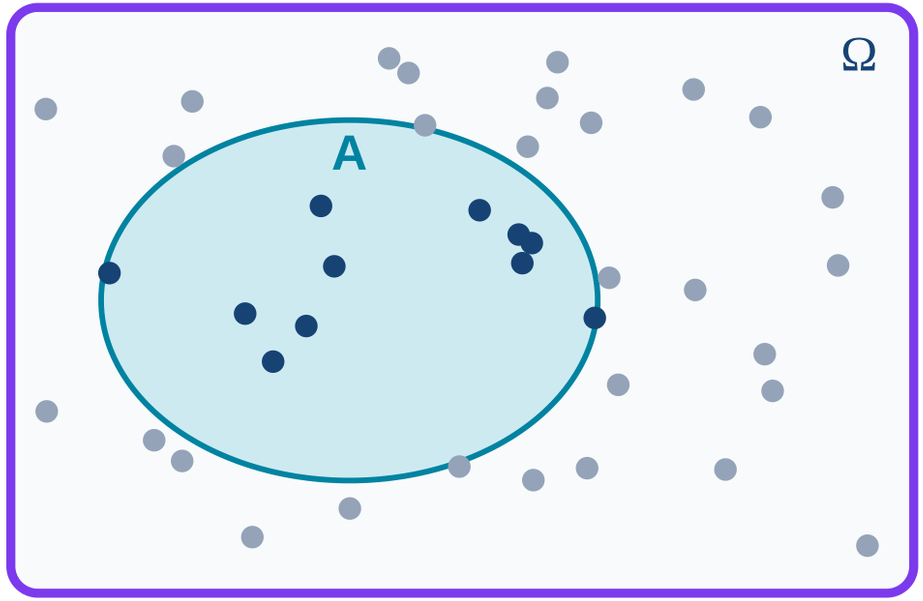
Module 3

What Is a Probability?

> ***Guiding Question:*** *Every attempt to define probability turns out to be circular. What do we do instead?*

An Intuition: Probability as Area

Throw 40 stones at random into Ω



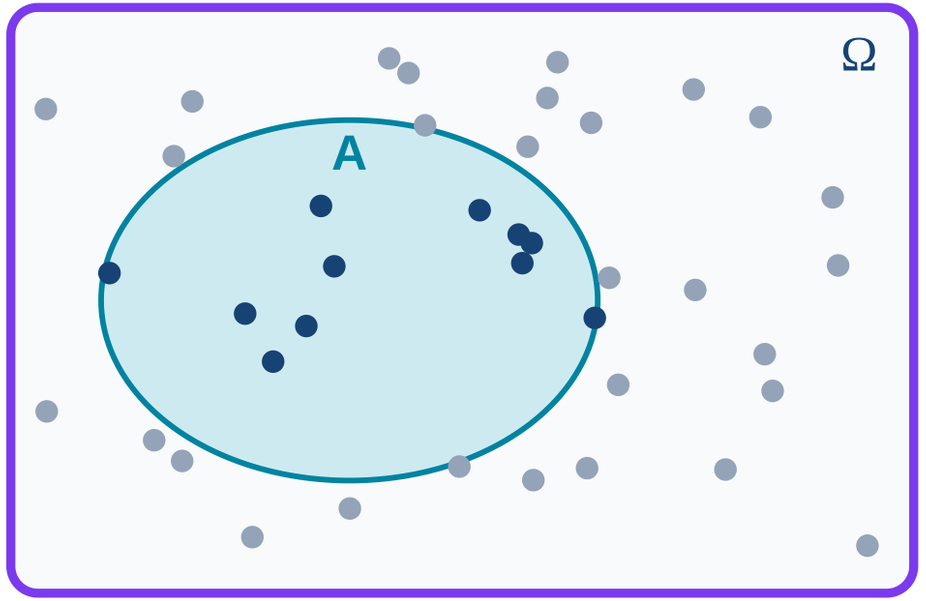
P(A) = area of A / area of Ω ≈ 11 of 40 stones

counted out of

Draw the sample space Ω on the ground, with each event as a region. Throw a stone at random.
How likely does it land in A?

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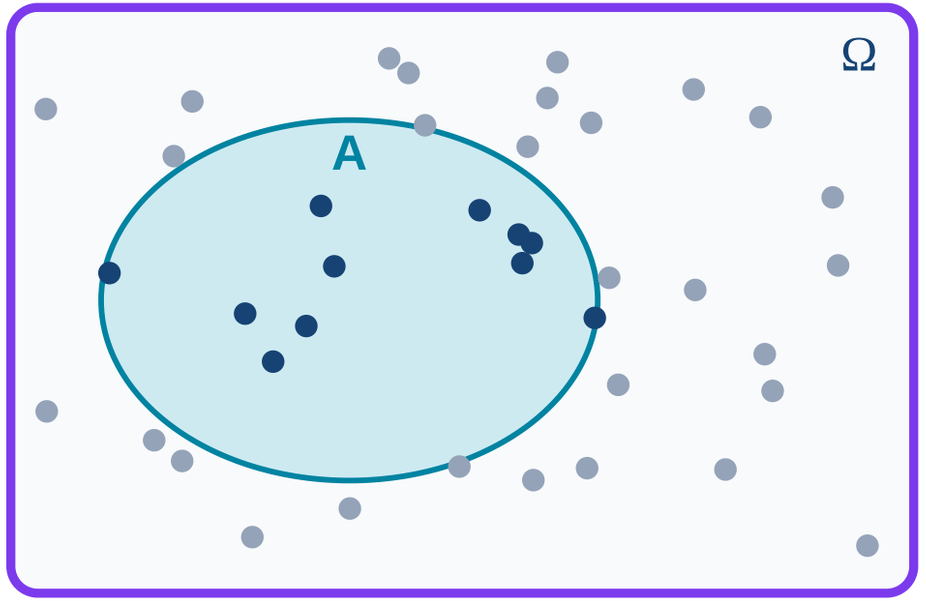
provided that

1. the stone always lands in Ω, and
2. every part of Ω is equally likely.

Probability measures **how much of** Ω an event occupies.

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Probability measures **how much of** Ω an event occupies.

> Assumption 2 is the one we will have to give up.

Two Ways to Read a Probability

Frequentist
The probability of an elementary event a is the fraction of repetitions of the experiment that yield a.
Requires the experiment to be repeatable.

Bayesian
The probability of a is the degree of certainty that a will happen — even without conducting the experiment.
Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment**; its probabilities need not be estimated, they are implied by the setup.

Why That Definition Is Not Enough

> **The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \to \mathbb{R}$ is a **probability measure** if

I	$P(A) \geq 0$	non-negative
II	$P(\Omega) = 1$	normalised
III	$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$	additive

INTUITION

Probability is **one kilogram of sand** spread over Ω; $P(A)$ is the sand lying in A.

- **I** — no region holds negative sand.
- **II** — all the sand lies somewhere in Ω.
- **III** — two regions that do not overlap hold their sand together.

The axioms say how *any* assignment must behave — not which probabilities the events get. How the sand is spread is the modeller's choice.

For infinite Ω the event space is a σ-algebra and Axiom III holds for countably many disjoint events — measure theory; nothing in this course depends on it.

Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

INTUITION

Three separate modelling choices: Ω — **what can happen** (the ground); $\mathcal{P}(\Omega)$ — **what may be asked** (the regions one may draw); P — **how likely each answer is** (how the sand is spread).

Same Ω, different P

A fair and a loaded die share $\Omega = \{1, \ldots, 6\}$ but are two probability spaces.

Finite Ω

P is fixed by the elementary events: $p_\omega \geq 0$, $\sum_\omega p_\omega = 1$, and $P(A) = \sum_{\omega \in A} p_\omega$.

What the Axioms Already Give Us

Property	Statement	From	Intuitively
Complement	$P(\overline{A}) = 1 - P(A)$	II, III	*the sand not in A lies in $\overline{A}$*
Empty event	$P(\emptyset) = 0$	complement, $A = \Omega$	*no region, no sand*
Bounds	$0 \leq P(A) \leq 1$	I, complement	*no region holds more than all the sand*
Monotonicity	$A \subseteq B \Rightarrow P(A) \leq P(B)$	I, III	*a larger region holds at least as much*
Inclusion–exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	III	*the overlap was weighed twice*
Union bound	$P(A \cup B) \leq P(A) + P(B)$	I, inclusion–exclusion	*weighing twice never weighs less*
Finite additivity	$P(A_1 \cup \ldots \cup A_p) = \sum_i P(A_i)$ for disjoint A_i	III, repeated	*disjoint regions add up*

Takeaway

> Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Laplace Probabilities in Practice

When all outcomes in Ω are equally likely, computing $P(A)$ means **counting twice**: the favourable cases, and the possible ones.

$$P(A) = \frac{|A|}{|\Omega|}$$

The whole difficulty is choosing a sample space in which "equally likely" is actually true.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

Why ordered pairs? Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

- $|\Omega| = 6 \cdot 6 = 36$ ordered pairs
- $E = \{(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)\}$, so $|E| = 6$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Why ordered pairs? Because $(1,6)$ and $(6,1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Ordered or Unordered — Both Work

An urn holds **6 white** and **5 black** balls. Three are drawn. **What is the probability of exactly 1 white and 2 black?**

Variant 1 — order matters

$|\Omega| = 11 \cdot 10 \cdot 9 = 990$

Favourable: white-black-black, black-white-black, black-black-white $= 6 \cdot 5 \cdot 4 + 5 \cdot 6 \cdot 4 + 5 \cdot 4 \cdot 6 = 360$

$$\frac{360}{990} = \frac{4}{11}$$

Variant 2 — order ignored

$|\Omega| = \binom{11}{3} = 165$

Favourable: $\underbrace{\binom{6}{1}}_{\text{the white}} \cdot \underbrace{\binom{5}{2}}_{\text{the blacks}} = 6 \cdot 10 = 60$

$$\frac{60}{165} = \frac{4}{11}$$

The General Rule

> When selecting k items from n, you may count **with** or **without** order — as long as you are consistent, and as long as the chosen sample space really is equiprobable.

Pick whichever counting is easier for the event at hand.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

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The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Unordered: $\binom{4}{2} = 6$ pairs, $2 \cdot 2 = 4$ mixed, so $4/6 = 2/3$. Ordered: $4 \cdot 3 = 12$ draws, $2 \cdot 2 + 2 \cdot 2 = 8$ mixed, so $8/12 = 2/3$ — the same, because both sample spaces are equiprobable.

Takeaway

Laplace's formula — favourable over possible cases — is **one** probability measure, valid only in a sample space whose outcomes are equally likely. The formula is trivial; choosing that sample space is the work.

Module 4

Conditioning, Independence, Bayes

> ***Guiding Question:*** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what is the correct bookkeeping?*

Motivating Example

You roll a 6-sided die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

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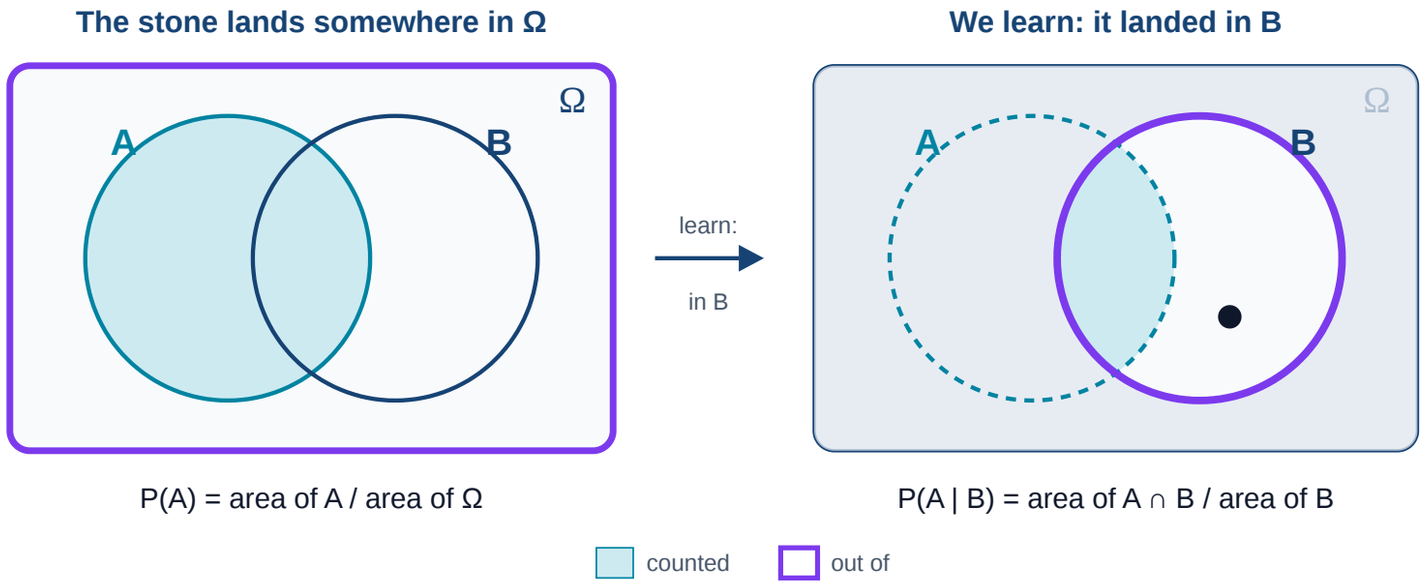
- Before the first roll: $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3,1),\ldots,(3,6)\}$ remains
- Outcomes like $(2,4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3,5)\}$

$$P = \frac{1}{6}$$

Definition

Definition (Conditional probability of A under the condition B).

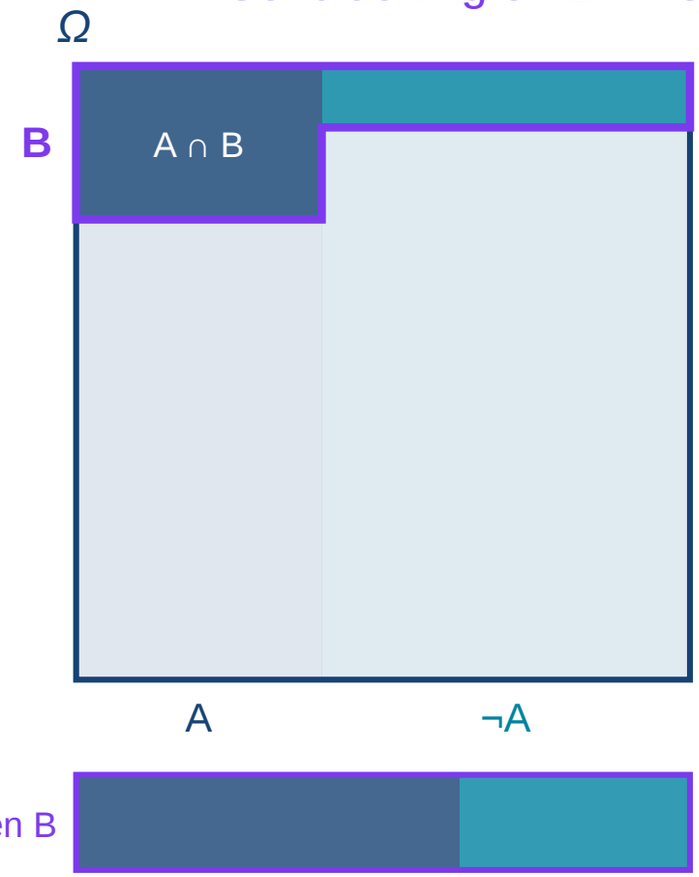
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{if } P(B) > 0$$



Back to the stone: once it is known to lie in B, B **is the whole world**.

Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1



P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

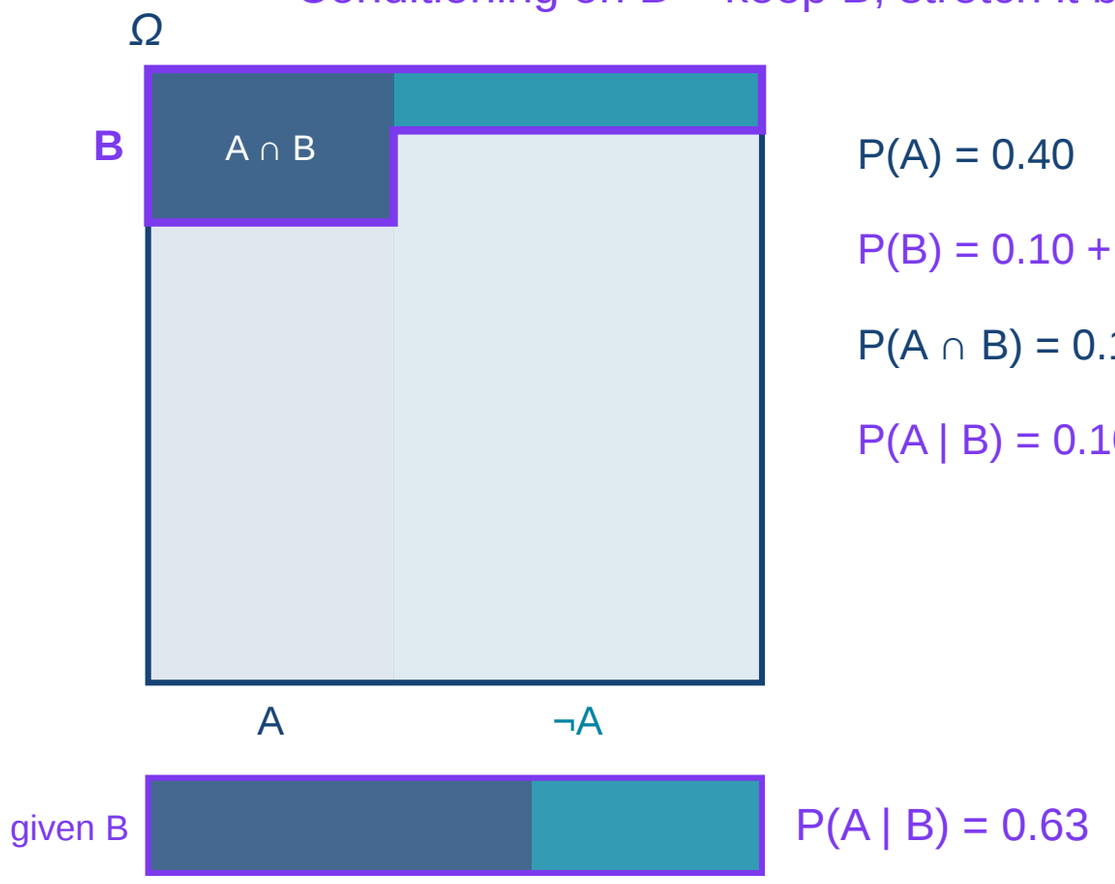
P(A | B) = 0.63

Before the last click: what share of B is A?

Then: set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

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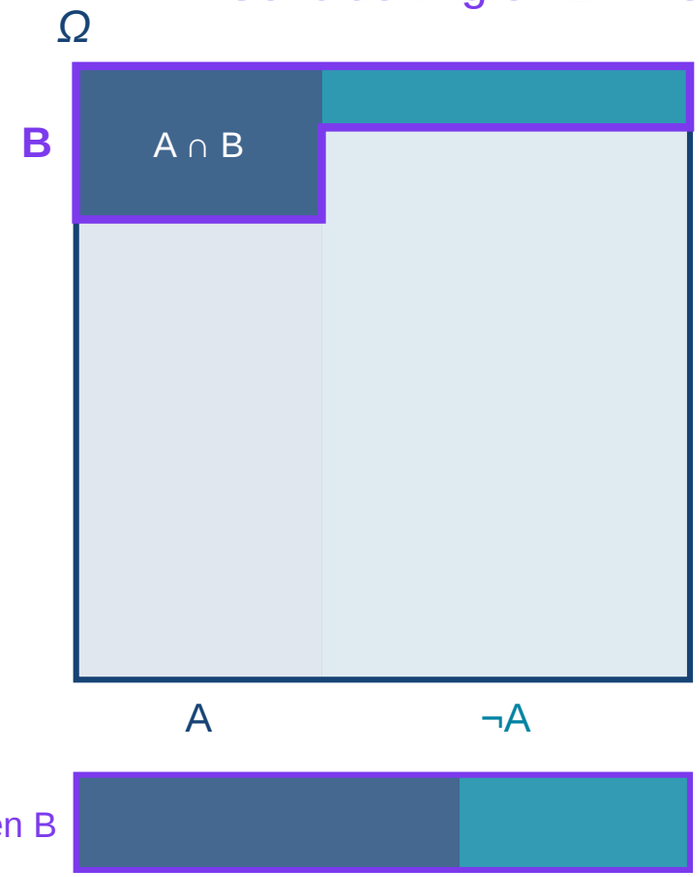
Before the last click: what share of B is A?

$P(A \cap B)/P(B) \approx 0.63$

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Then: set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

1: all of B lies inside A.

Example: Two Coins

Two fair coins are flipped. How likely is B = **both heads**, if we learn …

(a) "The first coin is heads"

$F = \{hh, ht\}$

(b) "At least one coin is heads"

$A = \{hh, ht, th\}$

Example: Two Coins

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(a) "The first coin is heads"

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$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

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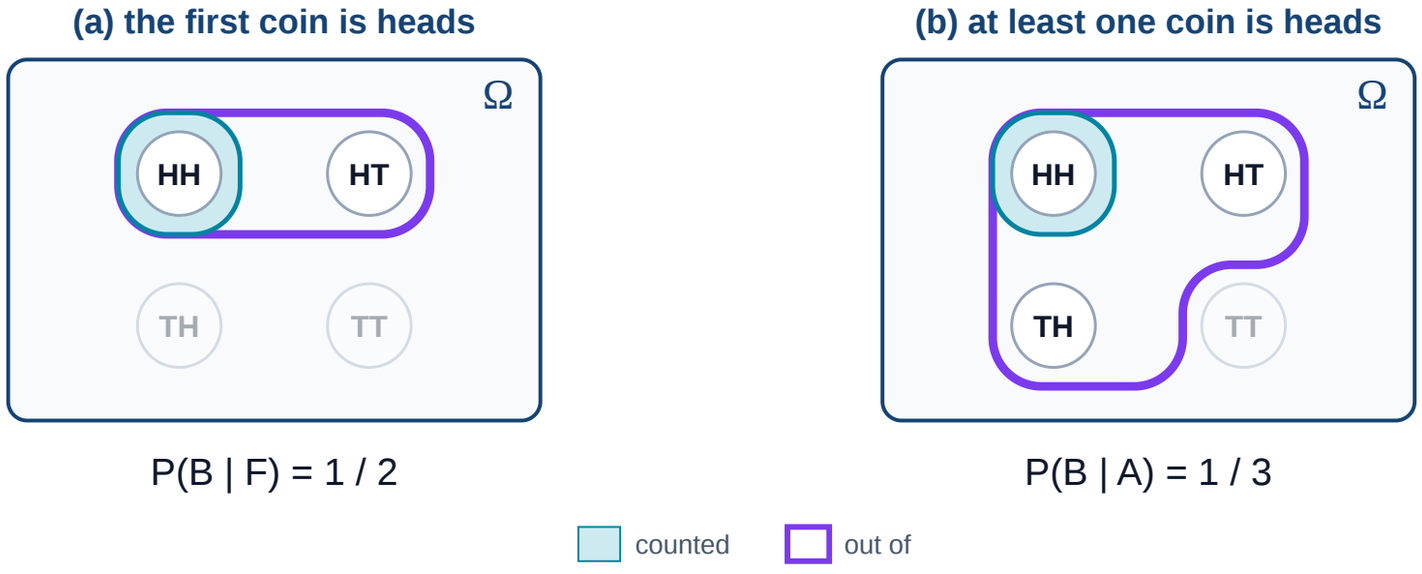
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$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

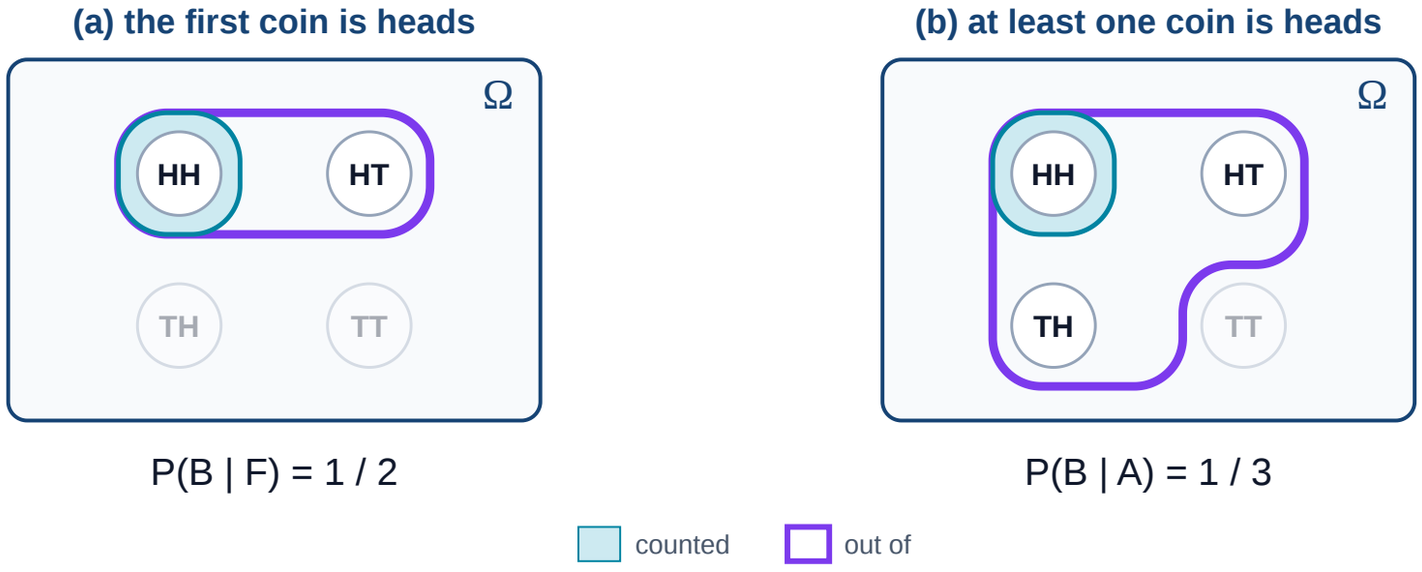
Same event, similar news. **Why is (b) smaller?**

Two Coins, Drawn



Both count the same outcome hh. News (a) leaves **two** possible outcomes, news (b) leaves **three**: weaker news rules out less, so hh gets a smaller share.

Two Coins, Drawn



Both count the same outcome hh. News (a) leaves **two** possible outcomes, news (b) leaves **three**: weaker news rules out less, so hh gets a smaller share.

> Pictures stop at two or three events. **Learn the rule** — it does not run out.

Total Probability

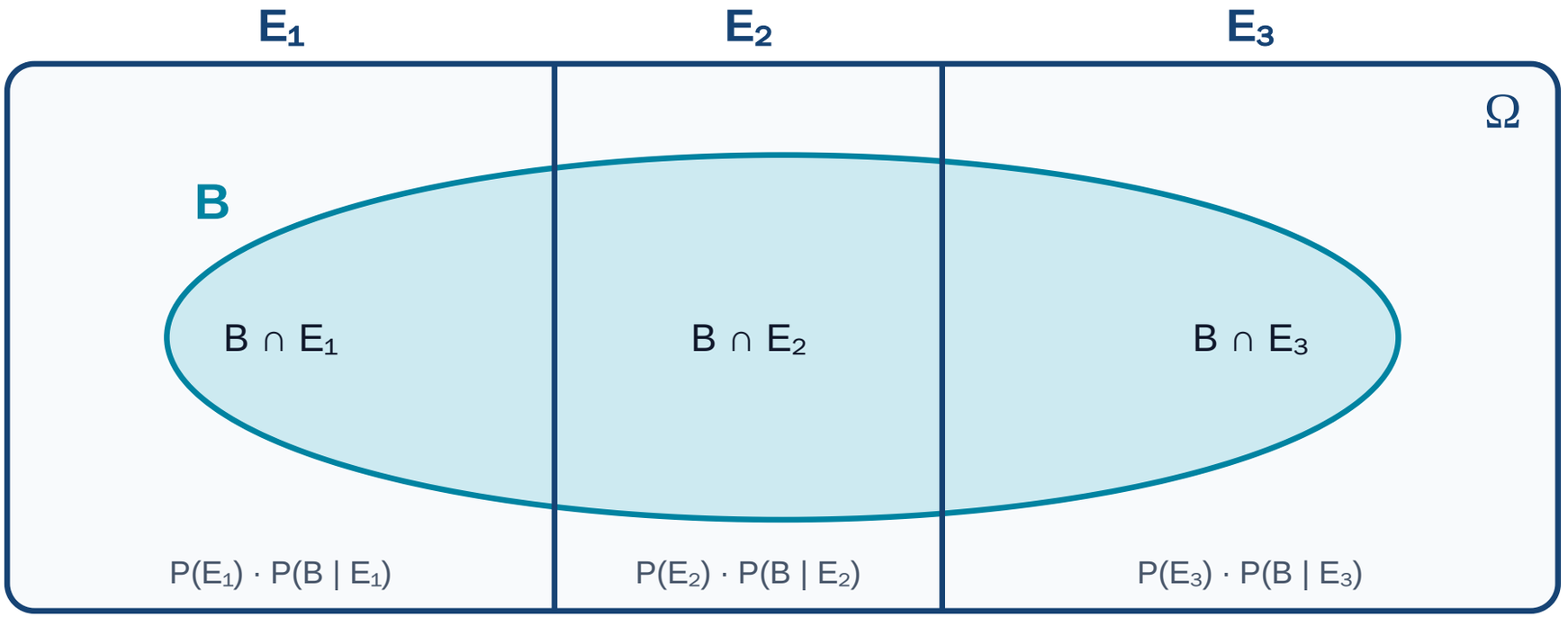
Theorem (Total probability). Let $E_1, \ldots, E_k$ be mutually exclusive with $\Omega = E_1 \cup \ldots \cup E_k$ and $P(E_i) > 0$. Then for any event B:

$$P(B) = \sum_{i=1}^{k} P(E_i) \cdot P(B \mid E_i)$$

Proof. Decompose B along the partition and use additivity:

$$P(B) = P((E_1 \cup \ldots \cup E_k) \cap B) = \sum_{i=1}^{k} P(E_i \cap B) = \sum_{i=1}^{k} P(E_i) \cdot P(B \mid E_i)$$

Total Probability, Drawn



$$P(B) = P(B \cap E_1) + P(B \cap E_2) + P(B \cap E_3) = \Sigma_i\, P(E_i) \cdot P(B \mid E_i)$$

B is cut into disjoint pieces, one per E_i; each piece is "first E_i, then B within E_i" — the product rule.

Rules That Follow

Product rule

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$
$$= P(A) \cdot P(B \mid A)$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

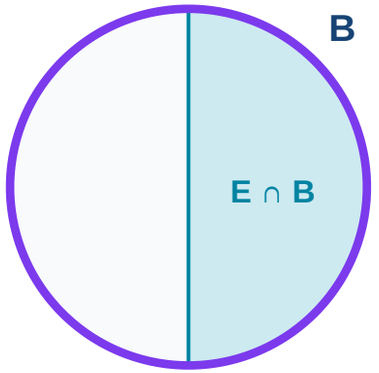
But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B, $P_B(E) = P(E \mid B)$ for every event E satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B.

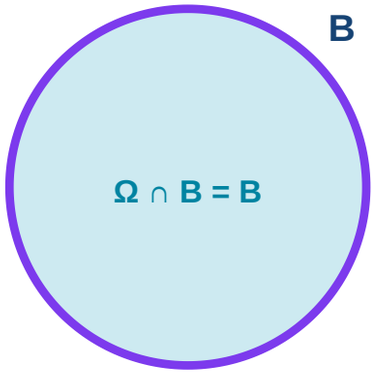
Conditioning Gives a New Measure

$P_B(E) = P(E \cap B) / P(B)$: every event E is measured by its piece inside B



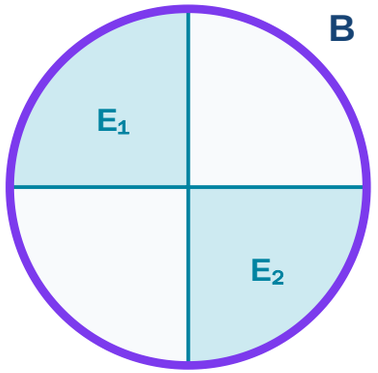
Axiom I: $P_B(E) \geq 0$

a piece of B is never negative



Axiom II: $P_B(\Omega) = 1$

all of B is the whole world



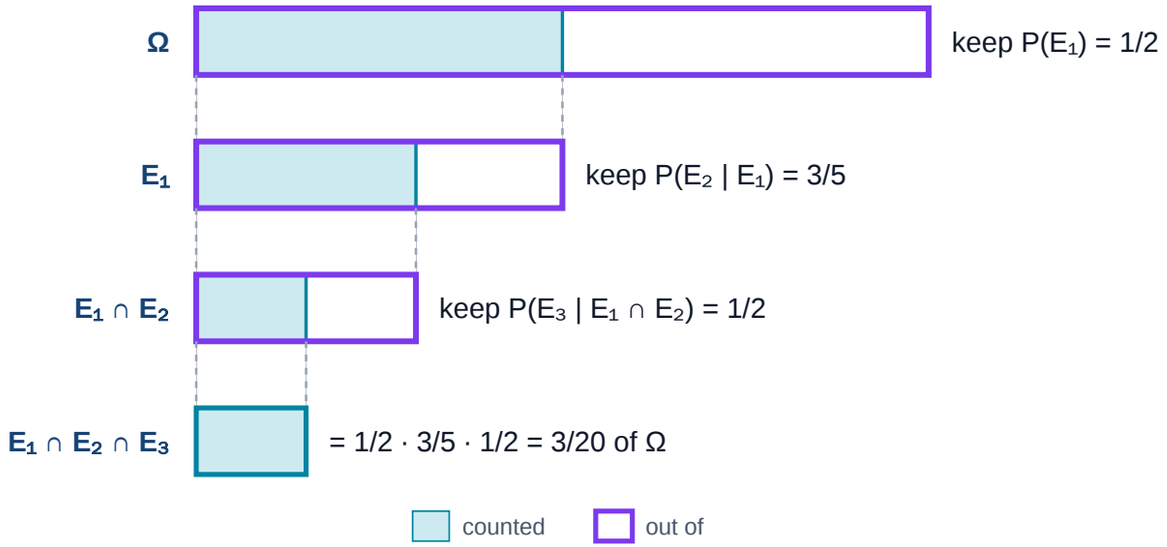
Axiom III: $P_B(E_1 \cup E_2) = P_B(E_1) + P_B(E_2)$

disjoint pieces of B add up

Inside B, the three axioms hold again — so **every rule proved for P holds for P_B**.

Chain Rule

$$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \ldots \cap E_{n-1})$$



Each factor keeps a share of what is left. Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Takeaway

> Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

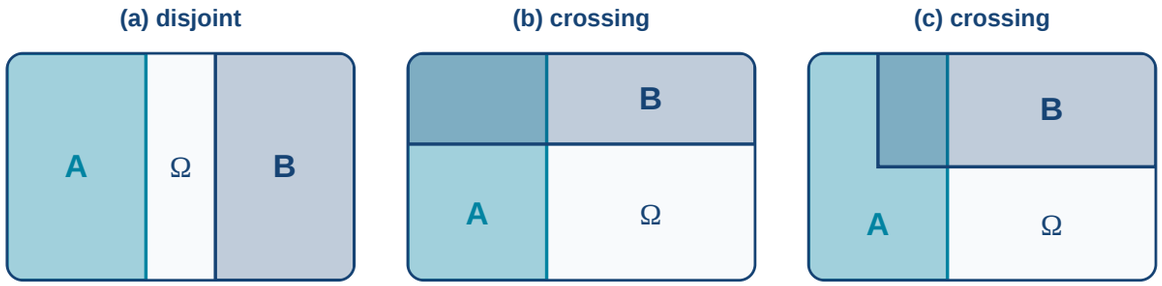
$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(A \mid B) = P(A \mid \overline{B})$$

Independence means: **learning B changes nothing about A.**

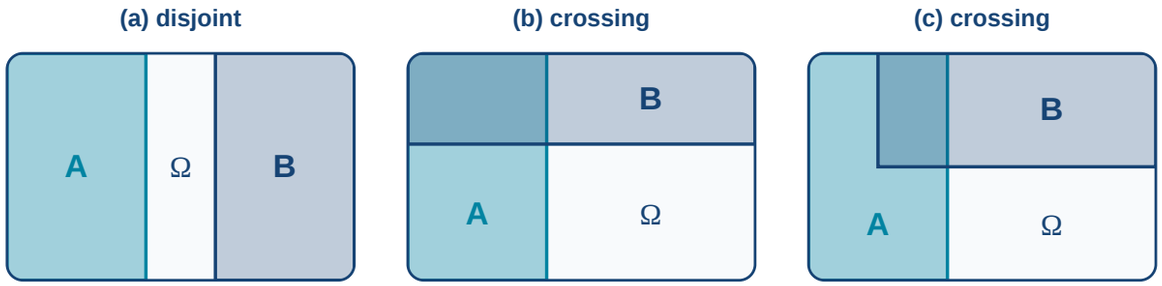
The Classic Confusion



In all three: P(A) = P(B) = 0.4

Which pair is **independent** — and why?

The Classic Confusion



In all three: P(A) = P(B) = 0.4

Which pair is **independent** — and why?

(a) dependent. If B occurs, A certainly did not: $P(A \mid B) = 0 \neq 0.4$.

(b) independent. A takes the same share of B as of Ω: $P(A \mid B) = 0.16/0.4 = 0.4$.

(c) dependent. They overlap, but too little: $P(A \mid B) = 0.1/0.4 = 0.25 \neq 0.4$.

> **Statistical independence does not mean the events are disjoint.** Disjoint events with positive probability are about as **dependent** as events can be.

Independence Is About Areas, Not Overlap

P(A) · P(B) = 0.30 · 0.40 = 0.12 = P(A ∩ B) → independent



P(A) = 0.30

P(B) = 0.40

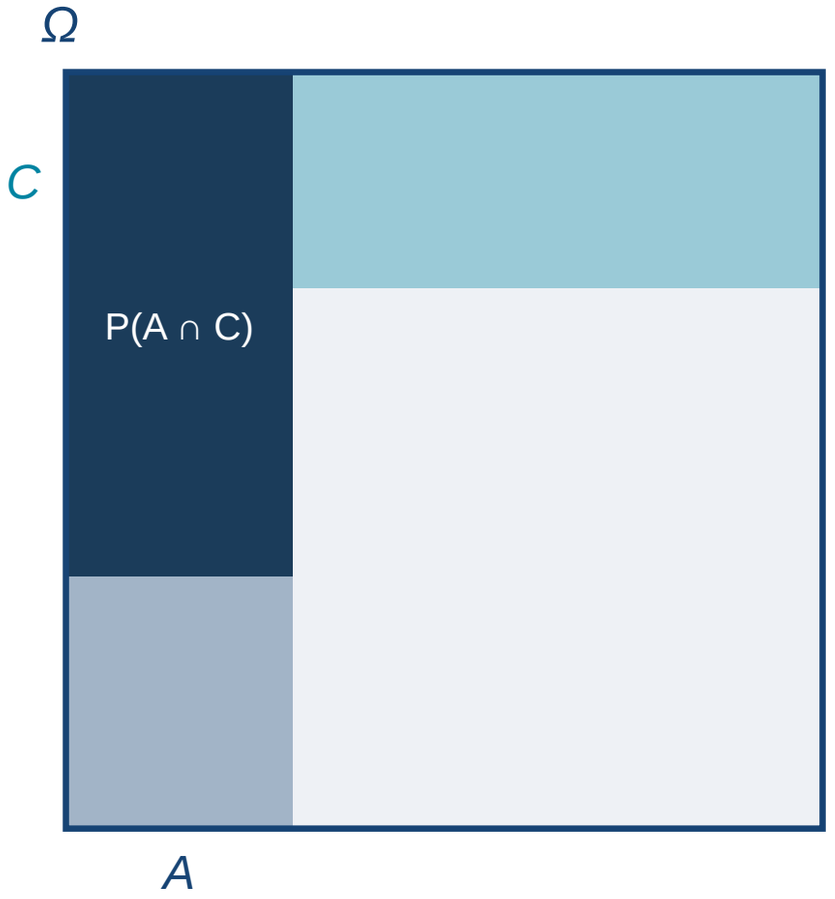
P(A ∩ B) = 0.12

P(A | B) = 0.12 / 0.40 = 0.30 = P(A)

Independent: **knowing B tells nothing about A.** A takes the same share of B as of all of Ω, so its probability stays the same.

The Same Picture, Dependent

P(A) · P(C) = 0.30 · 0.40 = 0.12 ≠ 0.20 = P(A ∩ C) → dependent



P(A) = 0.30

P(C) = 0.40

P(A ∩ C) = 0.20

P(A | C) = 0.20 / 0.40 = 0.50 ≠ P(A)

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

More Than Two Events

Events $A_1, \ldots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \ldots, i_l$:

$$P(A_{i_1} \cap \ldots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Takeaway

Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Conditional Independence

Definition. Events A and B are **conditionally independent given** C (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A.**

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$\frac{1}{16} \neq \frac{3}{32}$, so $A \not\perp\!\!\!\perp B$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

so $A \perp\!\!\!\perp B \mid C$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

Why This Matters

Common cause
A common cause C can make two effects look related; given C, they are independent.

Common effect
Two independent causes become dependent once their common effect is observed — *explaining away*.

Model size
Conditional independence lets a joint distribution factorise into small pieces — naive Bayes and graphical models are built on it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Takeaway

> Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Bayes' Theorem — An Intuition First

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Your test comes back positive. How likely is it that you are actually sick?

> Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

P(sick | positive) = 1/11 ≈ 9%



What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B; the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

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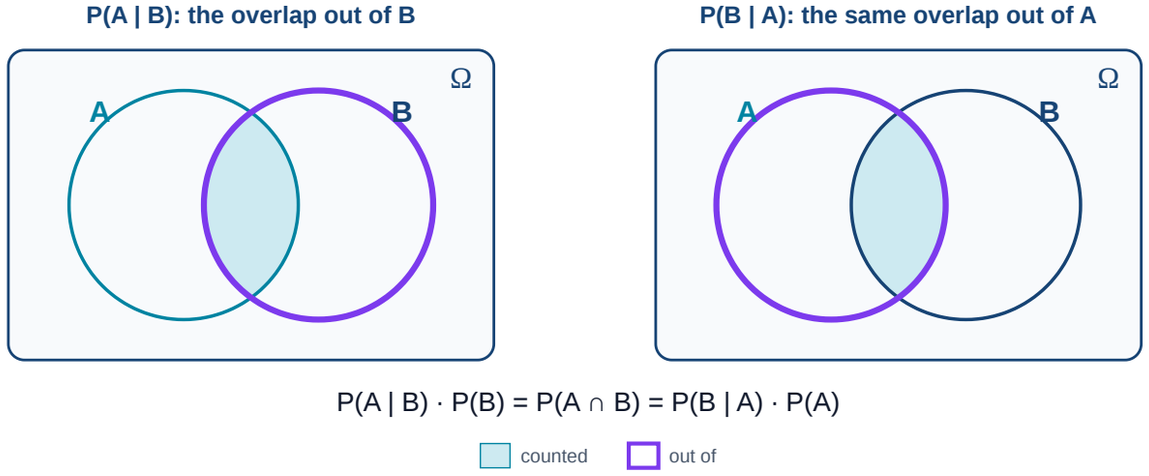
prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

"99% accurate" is the **likelihood** $P(+ \mid \text{sick})$; the intuition ignored the **prior** $P(\text{sick}) = 0.001$.

Proof — Two Events

Intuition



One overlap $A \cap B$, measured against two wholes: against B it is $P(A \mid B)$, against A it is $P(B \mid A)$.

Proof

step	because
$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$	definition
$= \frac{P(A \cap B)}{P(A)} \cdot \frac{P(A)}{P(B)}$	multiply and divide by $P(A)$
$= \frac{P(B \mid A)\, P(A)}{P(B)}$	definition, roles swapped

Needs $P(A), P(B) > 0$. ■

Mini-Exercise

A high-risk group. Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

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A high-risk group. Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

Mini-Exercise

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	everyone	high-risk group
the test, $P(+ \mid \text{sick})$ / $P(+ \mid \text{healthy})$	99% / 1%	99% / 1%
who is tested, $P(\text{sick})$	1 in 1,000	1 in 20
$P(\text{sick} \mid +)$	$\approx 9\%$	$\approx 84\%$

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who is tested, $P(\text{sick})$	1 in 1,000	1 in 20
$P(\text{sick} \mid +)$	$\approx 9\%$	$\approx 84\%$

> Both answers are right. Same test, same rule — **different people tested**.

The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

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$$= \frac{0.0495}{0.0495 + 0.0095} = \frac{0.0495}{0.0590} \approx 0.84$$

The denominator is the **evidence** $P(+)$: every way to test positive.

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$$= \frac{0.0495}{0.0495 + 0.0095} = \frac{0.0495}{0.0590} \approx 0.84$$

The denominator is the **evidence** $P(+)$: every way to test positive.

> Times 2,000, the fraction is **99 / 118**. Counting people and the formula are **one calculation**, written two ways.

Bayes' Theorem — Multiple Events

Theorem (Bayes, multiple events). Let $E_1, \ldots, E_k$ be mutually exclusive with $\Omega = E_1 \cup \ldots \cup E_k$ and $P(E_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(E_i \mid B) = \frac{P(B \mid E_i) \cdot P(E_i)}{P(B)} = \frac{P(B \mid E_i) \cdot P(E_i)}{\sum_{j=1}^{k} P(B \mid E_j) \cdot P(E_j)}$$

posterior of E_i

evidence

every E_i explains B, weighted by its prior

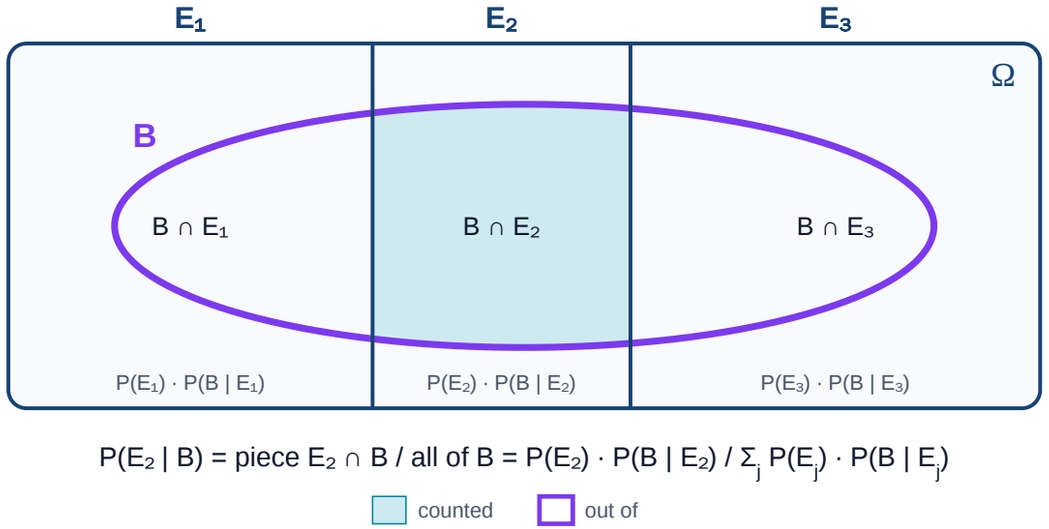
likelihood — how well E_i explains B

prior of E_i

The denominator is the **total probability** of the evidence.

Proof — Multiple Events

Intuition



The events E_i cut B into pieces. The posterior of E_i is **its piece out of all pieces**.

Proof

step	because
$P(E_i \mid B) = \frac{P(B\|E_i)\,P(E_i)}{P(B)}$	Bayes, two events
$P(B) = \sum_j P(B \cap E_j)$	additivity
$= \sum_j P(B \mid E_j)\,P(E_j)$	product rule

Additivity holds because the E_j are disjoint and cover Ω. Insert $P(B)$ into the first line. ■

Takeaway

> Bayes' theorem is the arithmetic of changing your mind: **posterior ∝ likelihood × prior**. Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Module 5

Random Variables and Distributions

> ***Guiding Question:*** *We rarely care which outcome occurred, only what it was worth. How do we move from outcomes to quantities?*

Why Random Variables?

Three fair coins: what matters is rarely *which* sequence occurred, but the **number of heads** Y.

Y	outcomes ω with $Y(\omega) = k$	count	$P(Y = k)$
0	TTT	1	$\frac{1}{8}$
1	HTT · THT · TTH	3	$\frac{3}{8}$
2	HHT · HTH · THH	3	$\frac{3}{8}$
3	HHH	1	$\frac{1}{8}$

> Y compresses eight outcomes into four numbers — the probabilities come along.

Definition

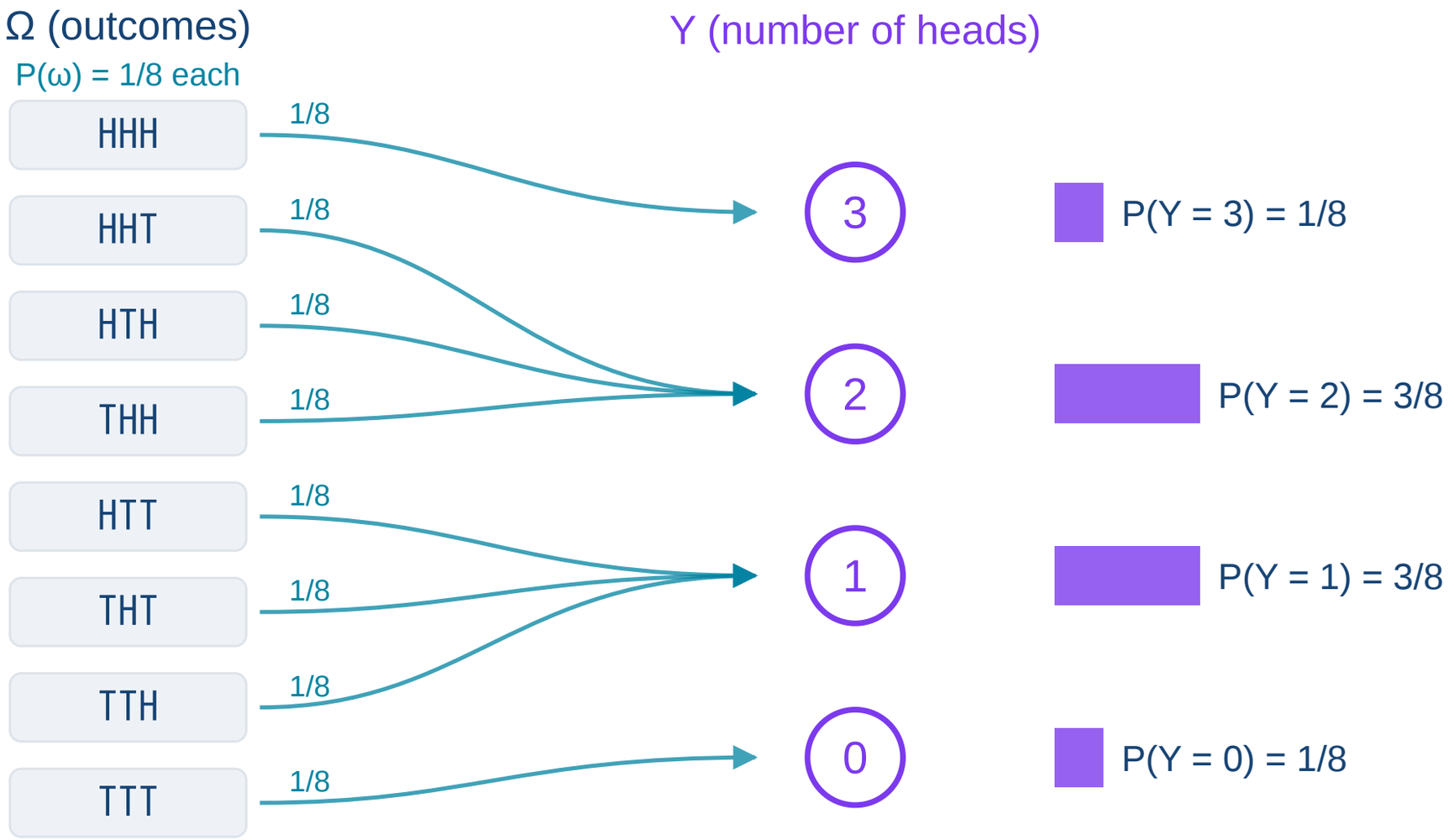
Definition (Random variable). A random variable X maps outcomes to a measurable range E: $X : \Omega \to E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω, not in X.

The Function, Drawn

Three arrows arrive at 1 and at 2: 3/8 each — the whole shape



Examples

experiment	sample space Ω	random variable $X(\omega)$	values
two coins	$\{HH, HT, TH, TT\}$	number of heads	$0, 1, 2$
two dice	pairs $\langle n_1, n_2 \rangle$	sum $n_1 + n_2$	$2, \ldots, 12$
draw a person	a population	height of the person	an interval of $\mathbb{R}$

The same outcome can carry many random variables: one per question asked of it.

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — with a mass function or with a density.

A Random Variable With Infinitely Many Values

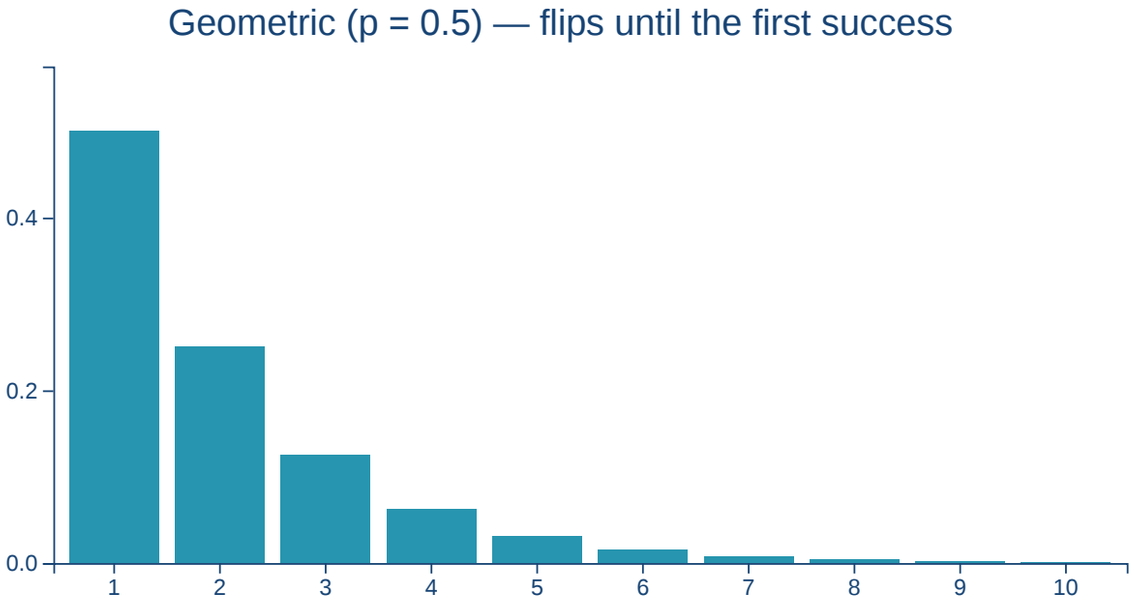
Flip a biased coin ($P(H) = p$) until the first head; X is the number of flips needed.

Geometric distribution, $X \sim \mathrm{Geom}(p)$:

$$P(X = 1) = p$$

$$P(X = 2) = (1 - p)p$$

$$P(X = n) = (1 - p)^{n-1}p$$



Countably infinite range, and the probabilities still sum to 1.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X, the **probability mass function** is

$$p(a) = P(X = a)$$

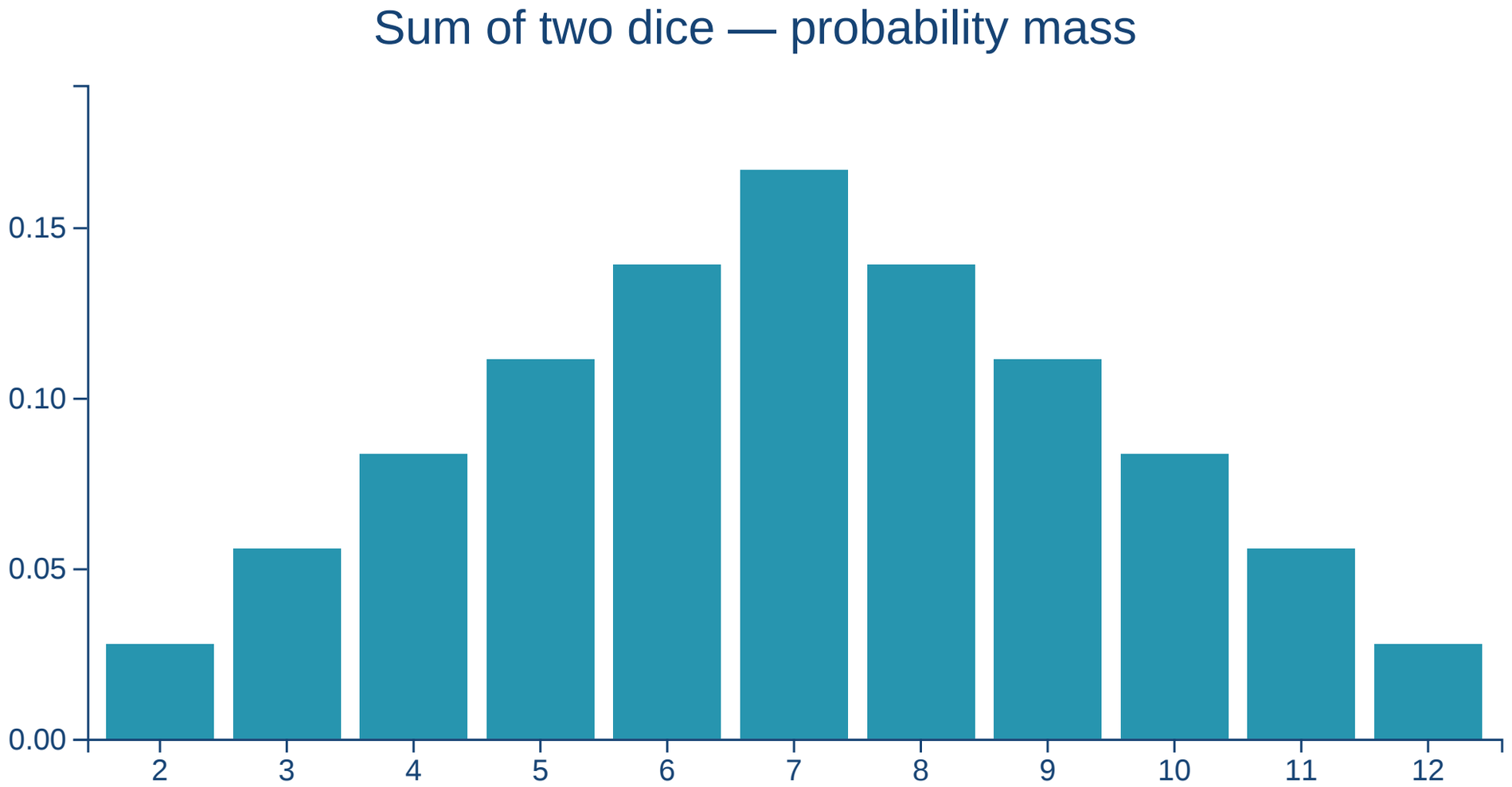
INTUITION

$p(a)$ is how much of the total probability sits exactly on the value a.

What follows from the axioms, with nothing new assumed:

- $p(a) > 0$ for at most countably many values a
- $\sum_i p(a_i) = 1$ — the masses add up to the whole
- p defines a probability measure on the values: the **distribution** of X

Example: Sum of Two Dice



The triangular shape is counting, drawn: six of the 36 outcomes give 7, only one gives 2.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

INTUITION

The **average outcome** if the experiment were repeated infinitely often — the balance point of the bar chart of p.

function of X	$E[g(X)] = \sum_i g(a_i)p(a_i)$
linear transformation	$E[aX + b] = aE[X] + b$
first moment	$\mu = E[X]$; the first *central* moment $E[X - \mu] = 0$

Variance

Definition (Variance).

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$$

Spread

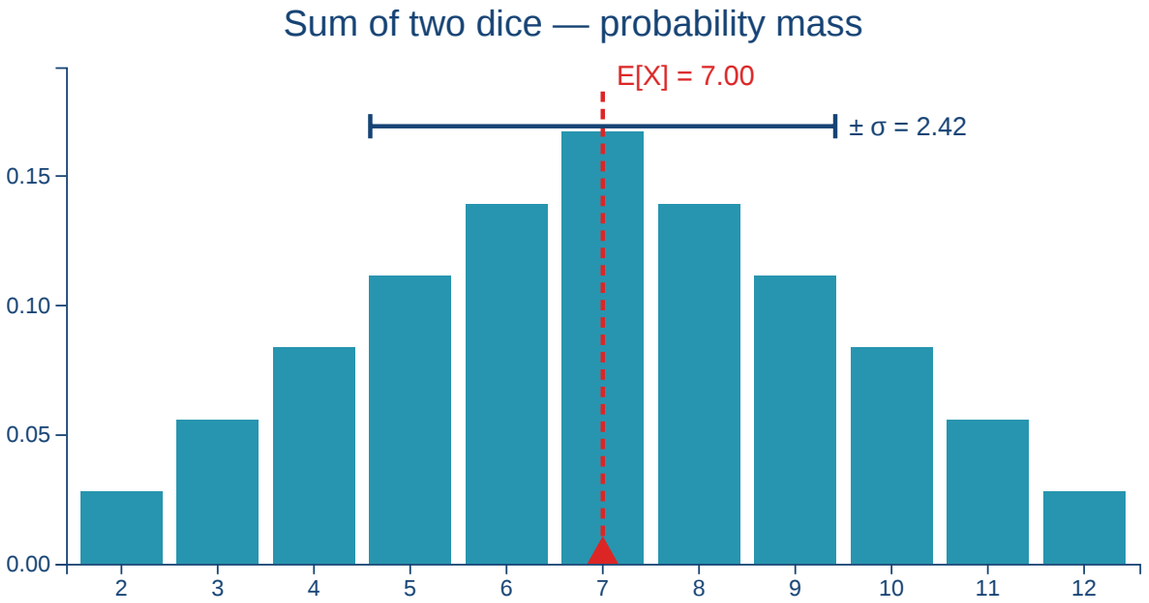
The second central moment: the expected squared distance from the mean. Larger variance, larger typical deviation.

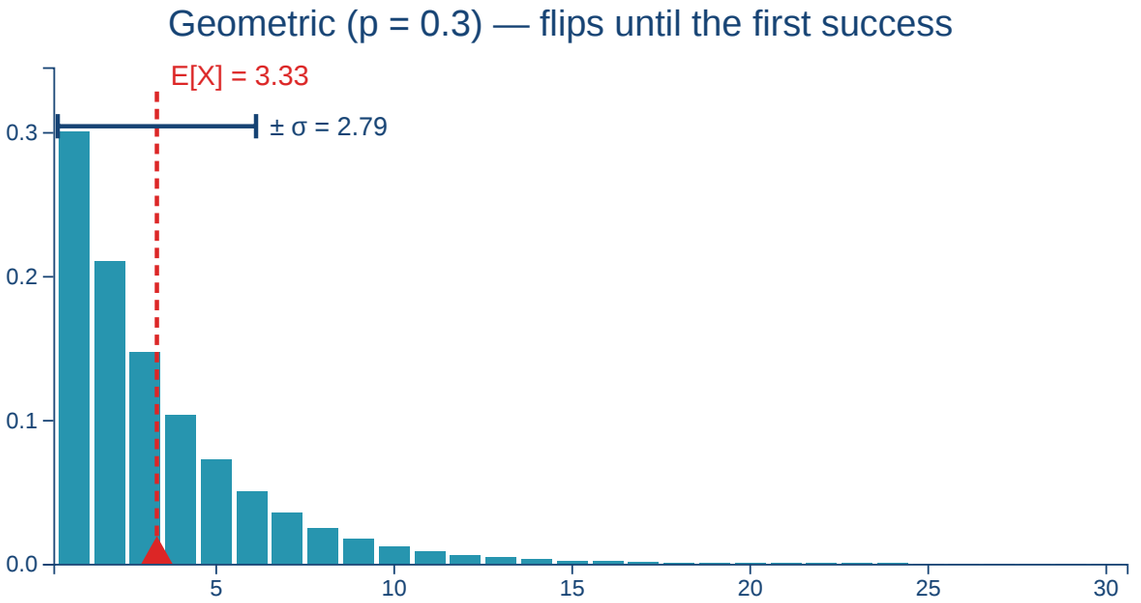
Scale and unit

$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$

$\sigma = \sqrt{\mathrm{Var}(X)}$, the **standard deviation**, has the unit of X.

Expectation and Variance, Drawn

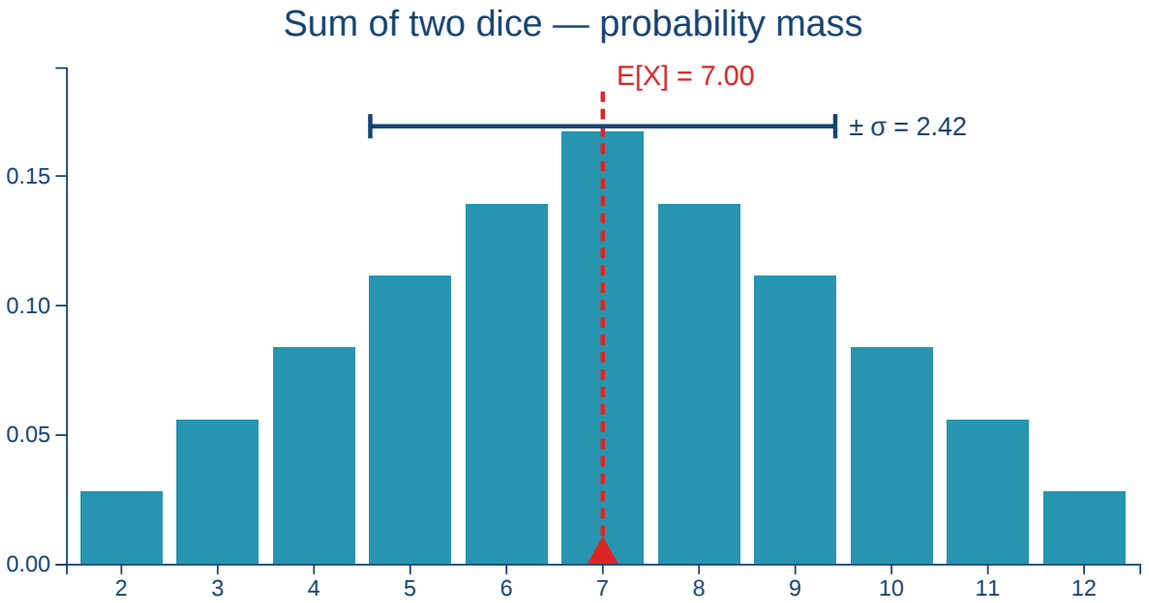


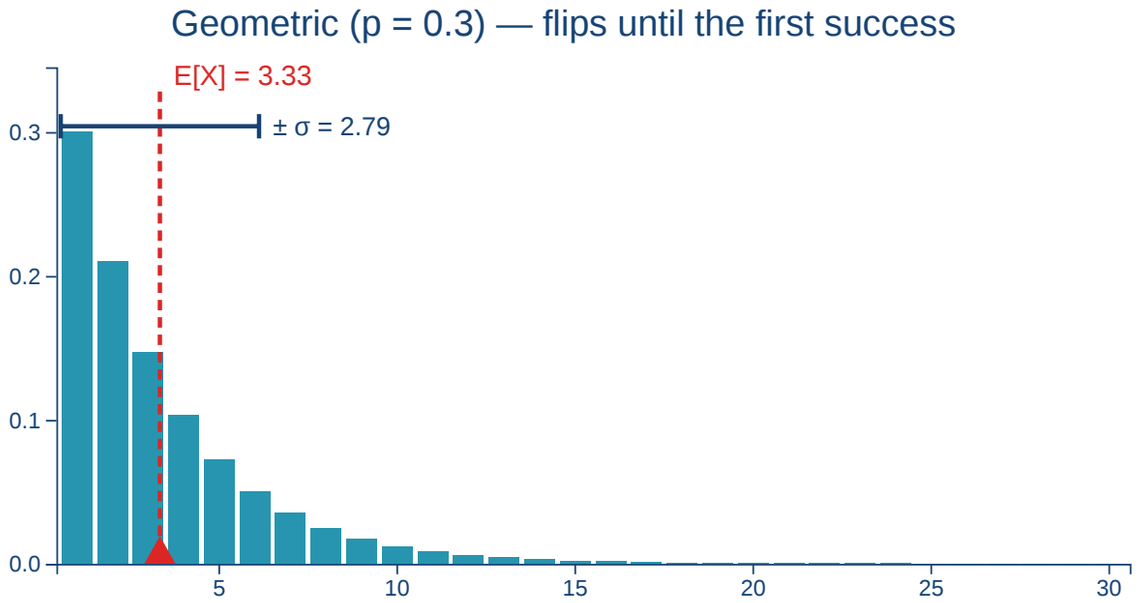


The red triangle marks $E[X]$: read the bars as weights on a beam, and the beam would rest level on it.

Does $E[X]$ sit under the **highest bar**?

Expectation and Variance, Drawn





The red triangle marks $E[X]$: read the bars as weights on a beam, and the beam would rest level on it.

Does $E[X]$ sit under the **highest bar**?

Only for the symmetric one. The geometric's highest bar is at 1, but $E[X] = 1/p \approx 3.33$: its **long right tail** pulls the expectation away from the peak, and widens σ.

Skewness

Definition (Skewness). With $\mu = E[X]$ and $\sigma^2 = \mathrm{Var}(X)$,

$$\mathrm{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{E\big[(X-\mu)^3\big]}{\sigma^3}$$

Cubing **keeps the sign** and lets far values dominate: a long right tail gives $\mathrm{Skew} > 0$, a long left tail $\mathrm{Skew} < 0$.

distribution	skewness	shape
sum of two dice	0	symmetric about 7
geometric, $p = 0.3$	$\frac{2-p}{\sqrt{1-p}} \approx 2.03$	long right tail

Takeaway

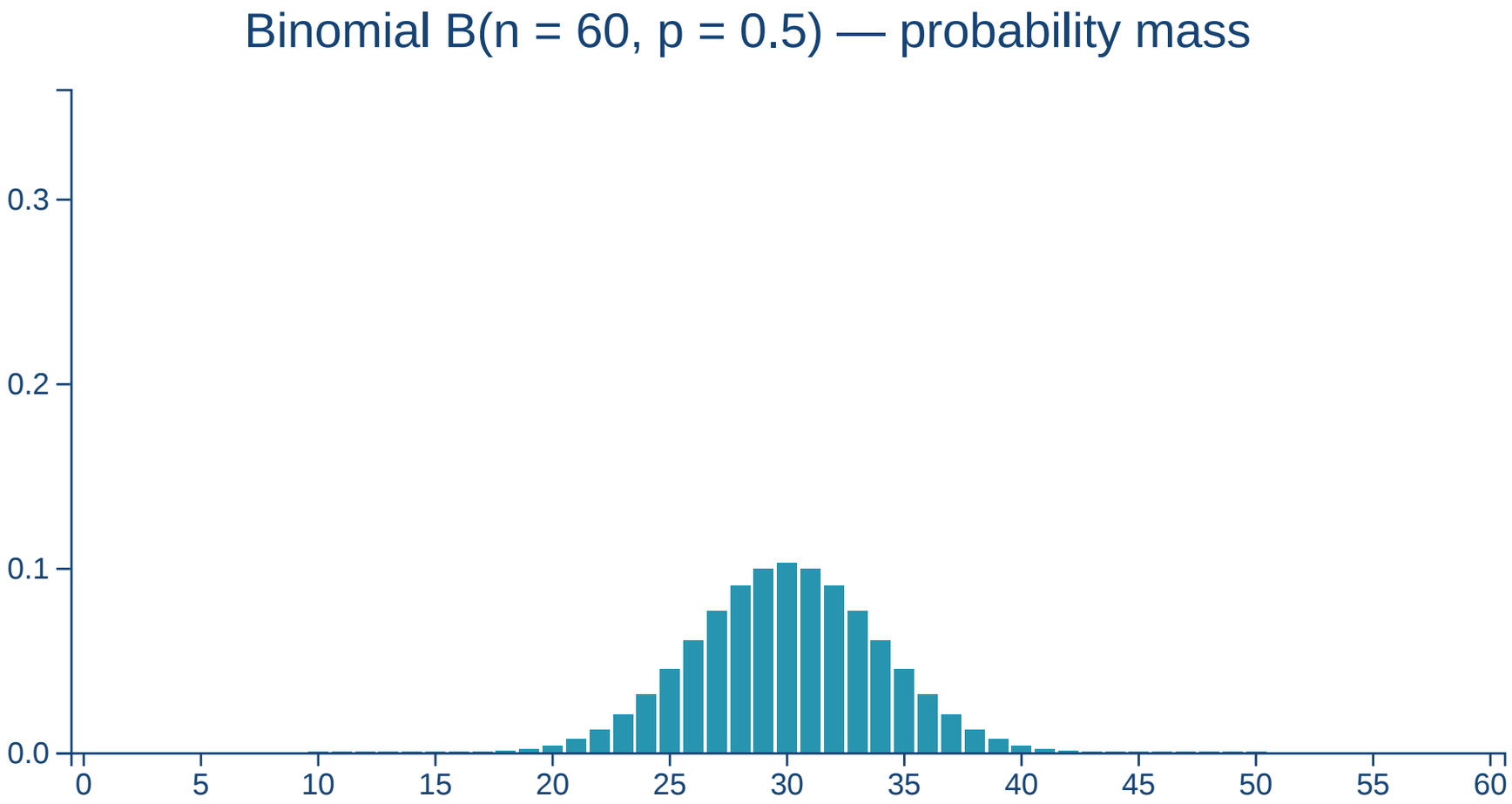
A discrete distribution is a PMF; expectation and variance compress it into **where it sits** and **how much it spreads**.

Bernoulli and Binomial

	models	PMF	$E[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	one trial: $X = 1$ success, $X = 0$ failure	$p(x) = p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial $B(n, p)$	successes in n independent trials	$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the sequences with k successes; the PMF multiplies it by the probability of one such sequence — combinatorics and independence in a single line.

How the Shape Depends on the Parameters



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

$B(10, 0.25)$: on average $10 \cdot 0.25 = 2.5$ correct. Five or more is twice the expectation — unlikely (about 8%).

Takeaway

A yes/no attribute is a **Bernoulli** variable with one parameter p; counting the yeses in n independent trials gives a **binomial**. Both are counting in probabilistic clothing — and p is the number data will have to tell us.

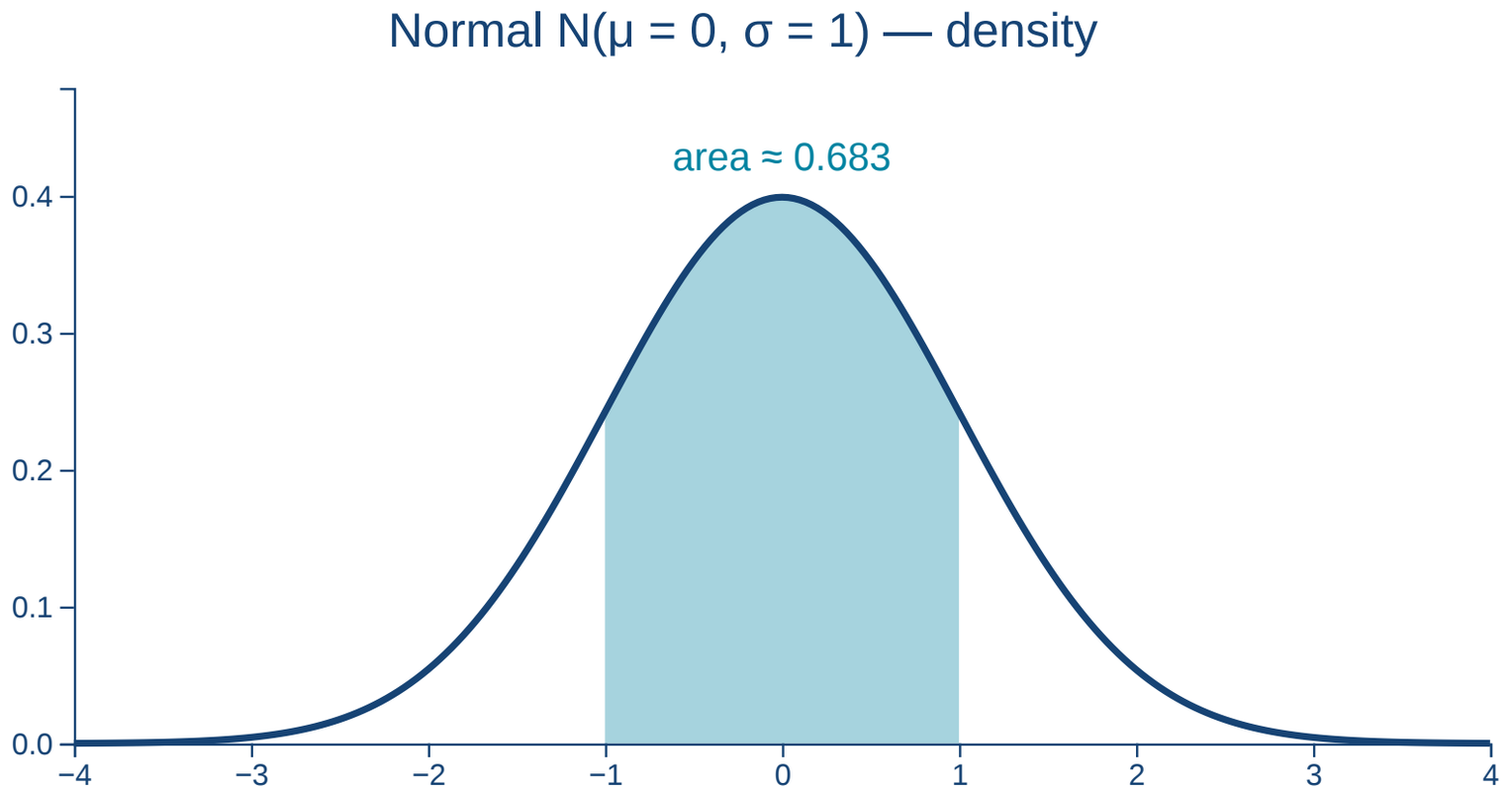
From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \leq X \leq c) = \int_b^c f(a)\,da, \qquad \int_{-\infty}^{\infty} f(a)\,da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a.

Probability Is Area Under the Curve



One value: probability 0. $P(X = a) = \int_a^a f = 0$ for every real a — a line has no area. The curve's height $f(a)$ is not a probability.

At 0 the curve is 0.40 high, yet $P(X = 0) = 0$.

The shaded region is $P(-1 \le X \le 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Expectation for Continuous Variables

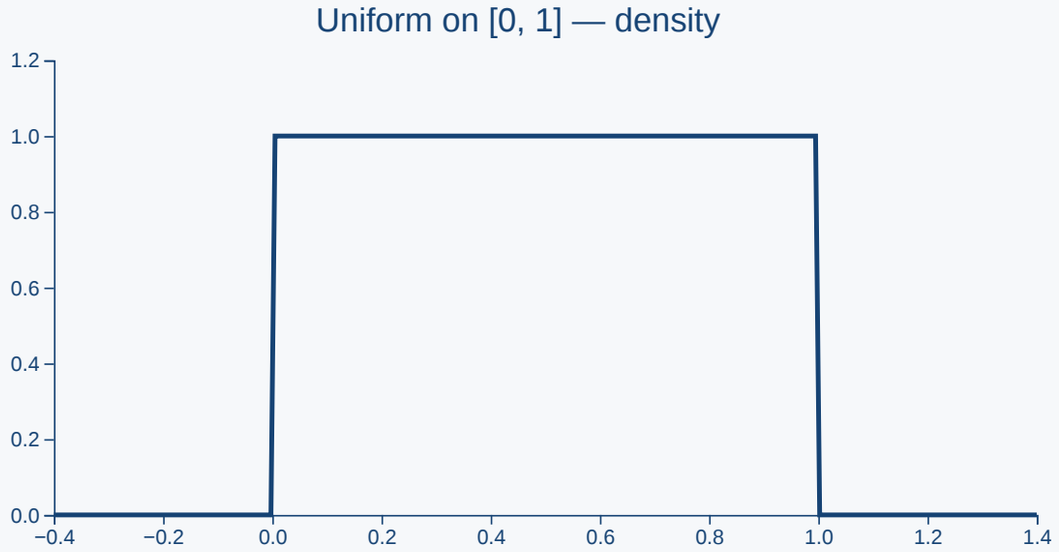
	discrete — mass p	continuous — density f
total	$\sum_i p(a_i) = 1$	$\int_{-\infty}^{\infty} f(a)\,da = 1$
expectation	$E[X] = \sum_i a_i p(a_i)$	$E[X] = \int_{-\infty}^{\infty} a f(a)\,da$
function of X	$E[g(X)] = \sum_i g(a_i) p(a_i)$	$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a)\,da$

Only the sum turns into an integral. The rules are unchanged: $E[aX + b] = aE[X] + b$ and $\mathrm{Var}(X) = E[(X - \mu)^2]$.

Two Basic Continuous Distributions

Uniform on $[a, b]$

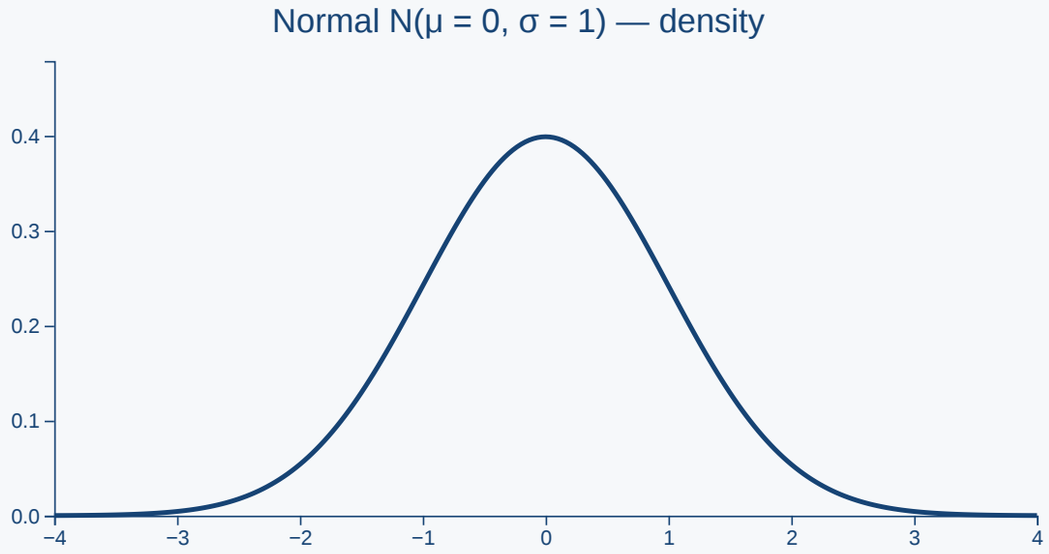
$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$



Equal-length intervals are equally likely — the Laplace assumption, continuous.

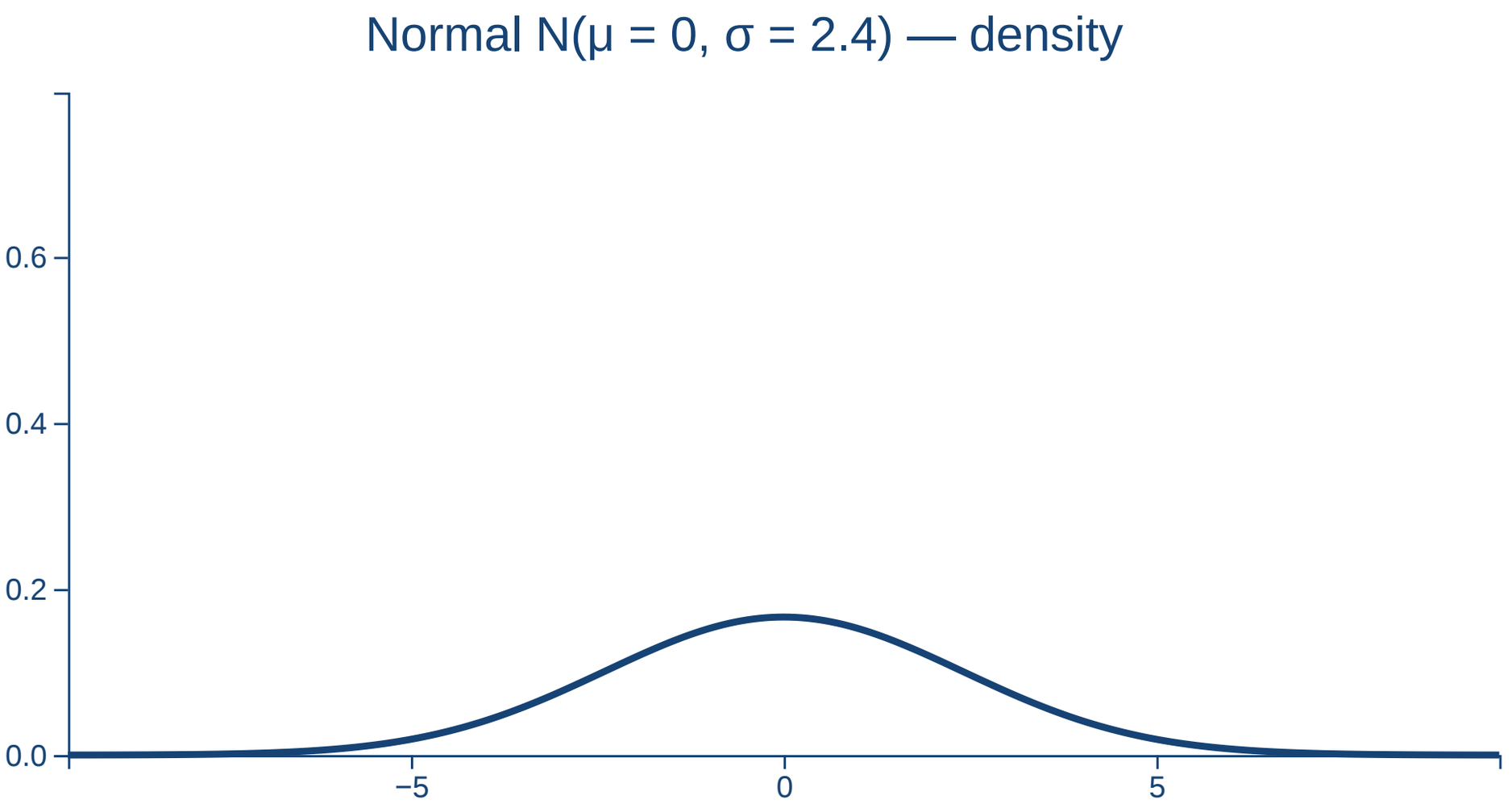
Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

What σ Does to the Shape



The peak falls as the curve widens — the area is always 1.

Why the Normal Is Everywhere

> The **central limit theorem**: under mild conditions, the average of n independent observations is approximately normal for large n — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown — and why a sum of many independent coin flips, the binomial, looks bell-shaped.

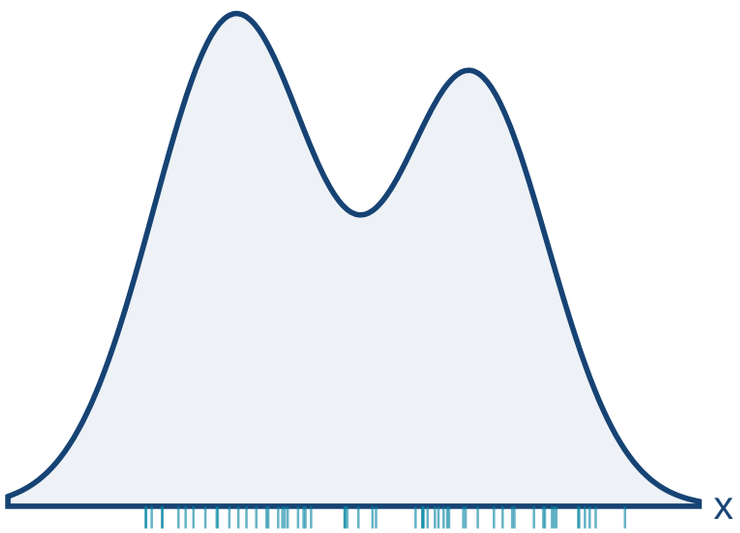
Other Distributions You Will Meet

distribution	values	typically models
Exponential	$[0, \infty)$	time until an event occurs (e.g. the next earthquake)
Gamma	$[0, \infty)$	time until n events have occurred
Beta	$[0, 1]$	an unknown probability
Dirichlet	probability vectors	distributions *over* probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

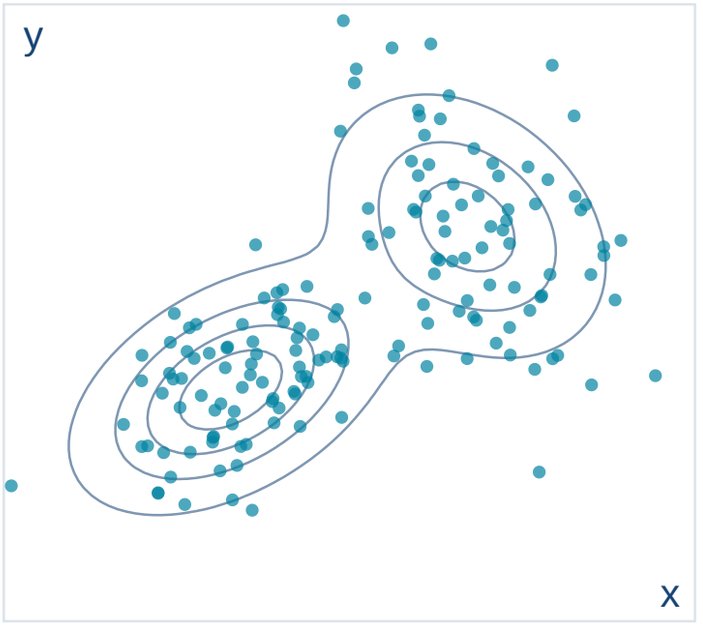
More Than One Random Variable

Univariate



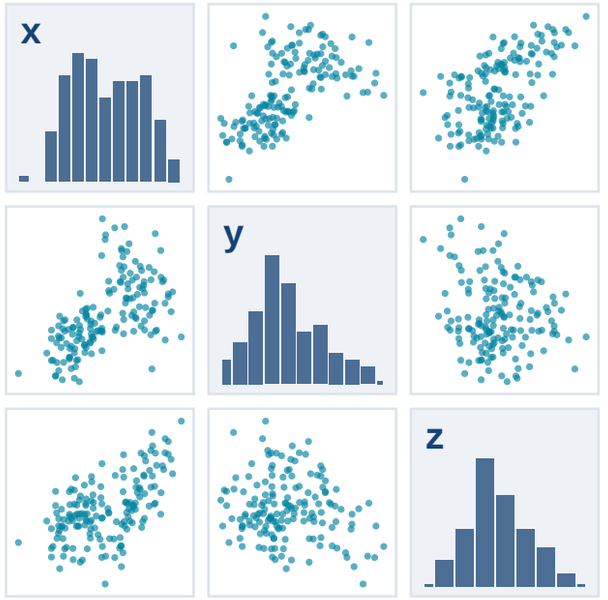
p(x) — one column

Bivariate



p(x, y) — two columns

Multivariate



p(x, y, z, …) — three or more

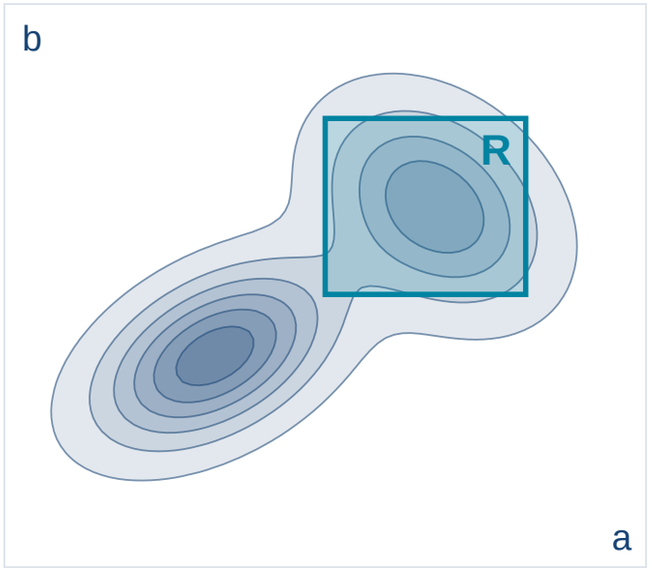
A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Joint Mass and Density

joint PMF (discrete)	joint PDF (continuous)
$p_{X,Y}(a,b) = P(X=a, Y=b)$	$P((X,Y)\in R) = \iint_R f_{X,Y}(a,b)\,da\,db$

	G=0	G=1	G=2	G=3
B=0	.15	.1	.0875	.0375
B=1	.1	.175	.1125	0
B=2	.0875	.1125	0	0
B=3	.0375	0	0	0

one probability per cell · all sum to 1



darker = denser · volume over R ≈ 0.33

From here on we write $p(a,b,c,\ldots)$ for the joint of $X_1, X_2, X_3, \ldots$ — a mass or a density, as the variables require.

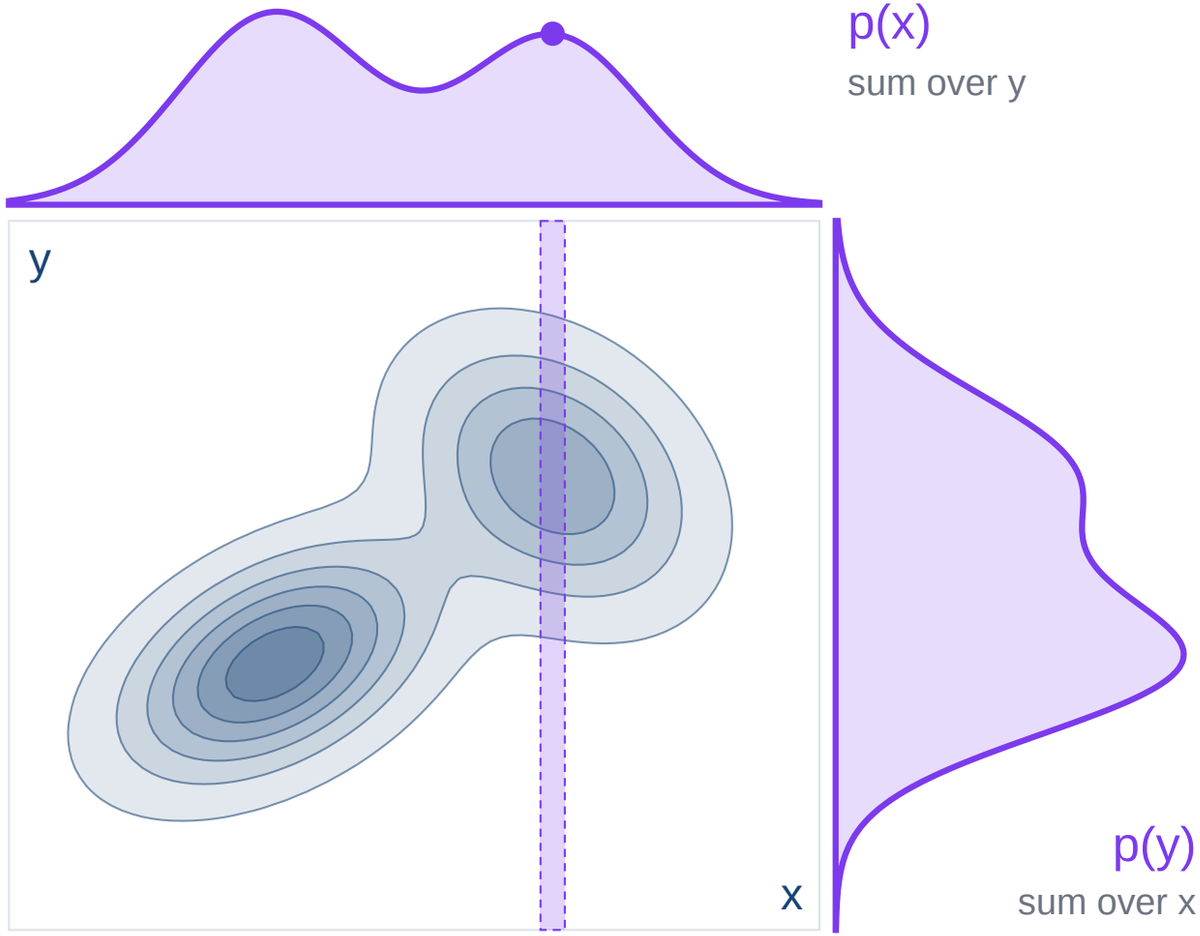
Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \ldots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \ldots, x_n)$$

INTUITION

Forget about a variable, but keep its influence correctly accounted for: collapse the joint onto the axis of the variable you keep.



Example: Boys and Girls

	G=0	G=1	G=2	G=3	p(B)
B=0	.15	.1	.0875	.0375	.375
B=1	.1	.175	.1125	0	.3875
B=2	.0875	.1125	0	0	.2
B=3	.0375	0	0	0	.0375
p(G)	.375	.3875	.2	.0375	total 1



Families: 15% have no children, 20% one, 35% two, 30% three. Each child is a boy or a girl with probability $\frac{1}{2}$, independently.

B boys, G girls.

> The margins of the table *are* the marginals — hence the name.

Covariance

$$\operatorname{cov}(A, B) = E\big[(A - E[A])(B - E[B])\big]$$

$\operatorname{cov} > 0$
A large when B large — positively related

$\operatorname{cov} < 0$
A large when B small — negatively related

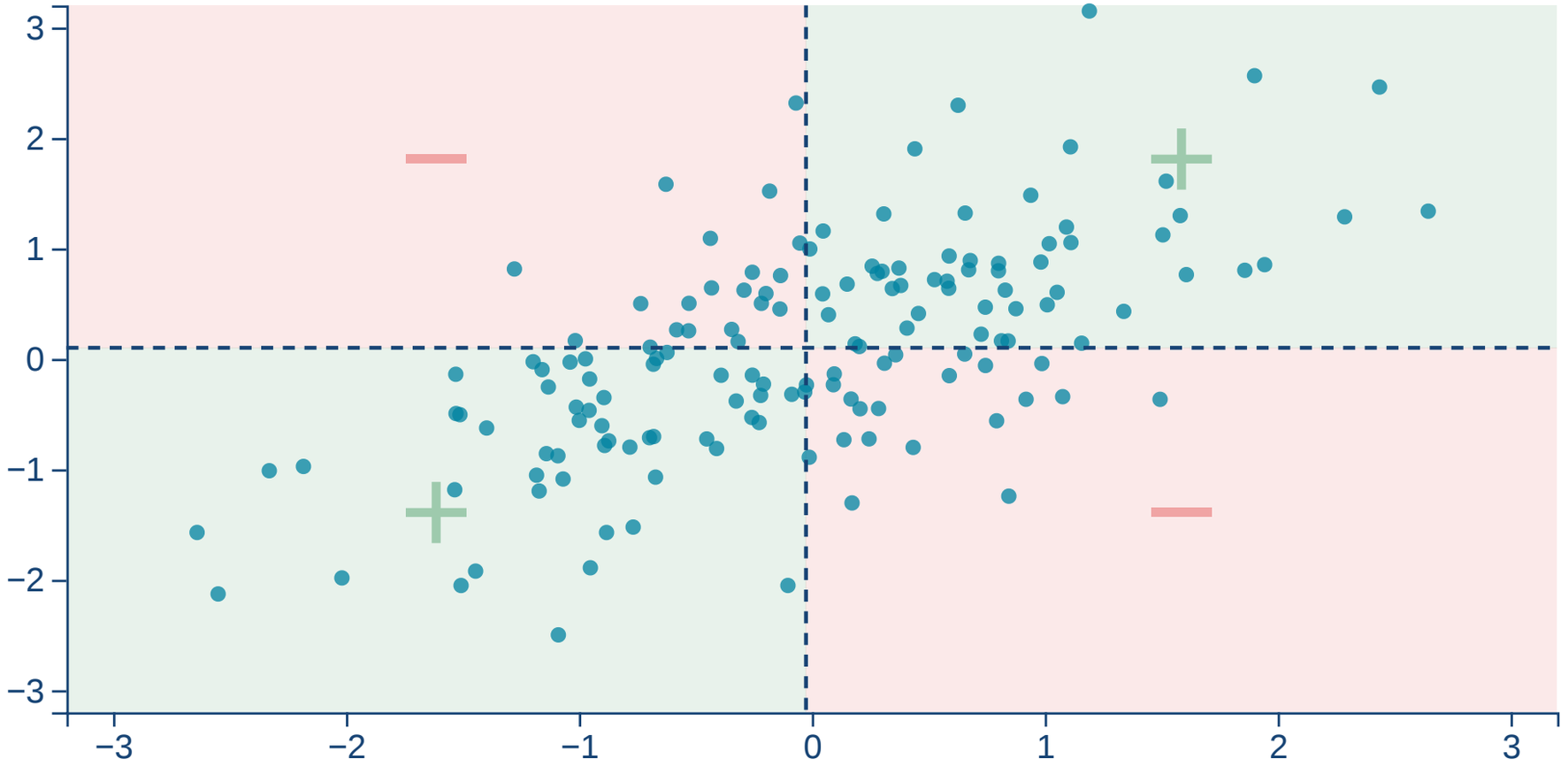
$\operatorname{cov}(A, A) = \operatorname{Var}(A)$
variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *linear* association.

Covariance Sees Only One Kind of Structure

Slide ρ, then switch the relation. Predict the covariance first.

cov(X, Y) = 0.62



Predict first: with $Y = X^2 - 1$, every point is fixed by its x. The covariance?

Takeaway

The joint distribution holds everything; **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 6

Describing Data

> ***Guiding Question:*** *The distribution is unknown and all we have is a sample. What can we honestly say about it?*

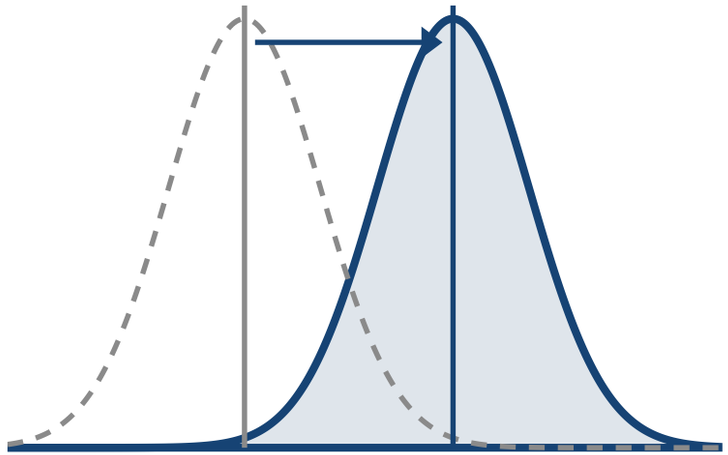
We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

	what it is	its numbers	do we see them?
Population	the entire group — all customers, all transactions, all documents	**parameters** μ, σ^2	no — the truth we want
Sample	the observations we actually have	**statistics** $\overline{x}$, s^2	yes — and we estimate the parameters from them

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

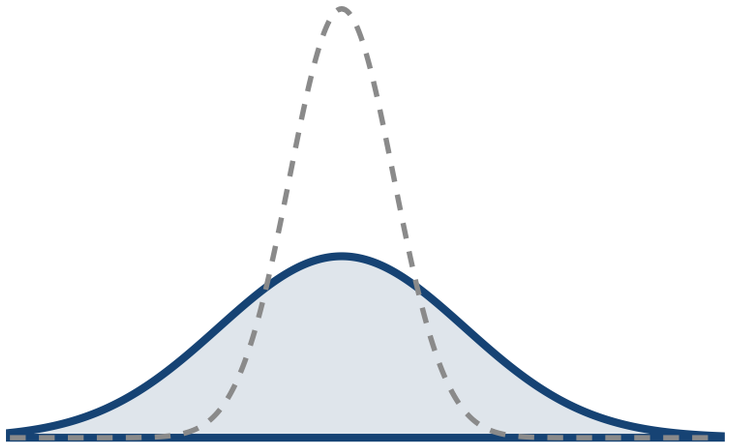
What We Summarise



Centre

1st moment

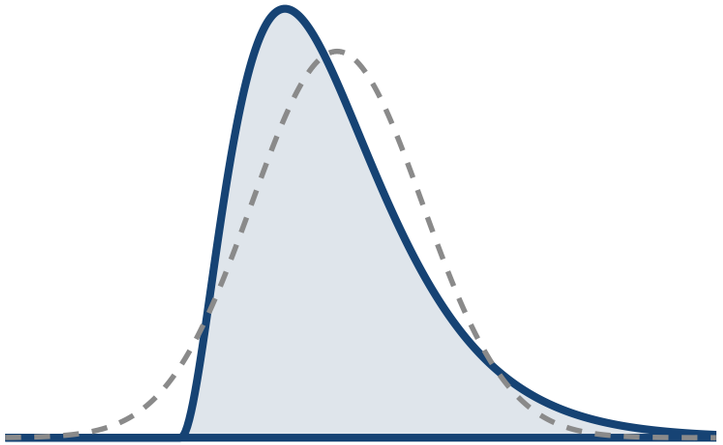
mean, median



Dispersion

2nd central moment

variance, std. dev., IQR



Shape

3rd central moment

skewness, quantiles

Univariate statistics describe one random variable; **multivariate** statistics describe $X_1, \ldots, X_d$ together.

Means

mean	formula	use it for	example
sample	$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	values that add up	average salary
weighted	$\overline{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$	values that count unequally	grade over courses of different credits
geometric	$\left(\prod_{i=1}^{n} x_i\right)^{1/n}$	factors that multiply	+50% then −50%: factor 0.87 per period
harmonic	$\frac{n}{\sum_i 1/x_i}$	rates over equal amounts	60 km/h out, 40 back: 48 km/h
trimmed	average after dropping the k smallest and k largest	robustness on the cheap	judges' scores

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$-th value, for even n the average of the two middle values.

INTUITION

Half of the values lie below it, half above. Only their **order** counts, not how far out they lie.

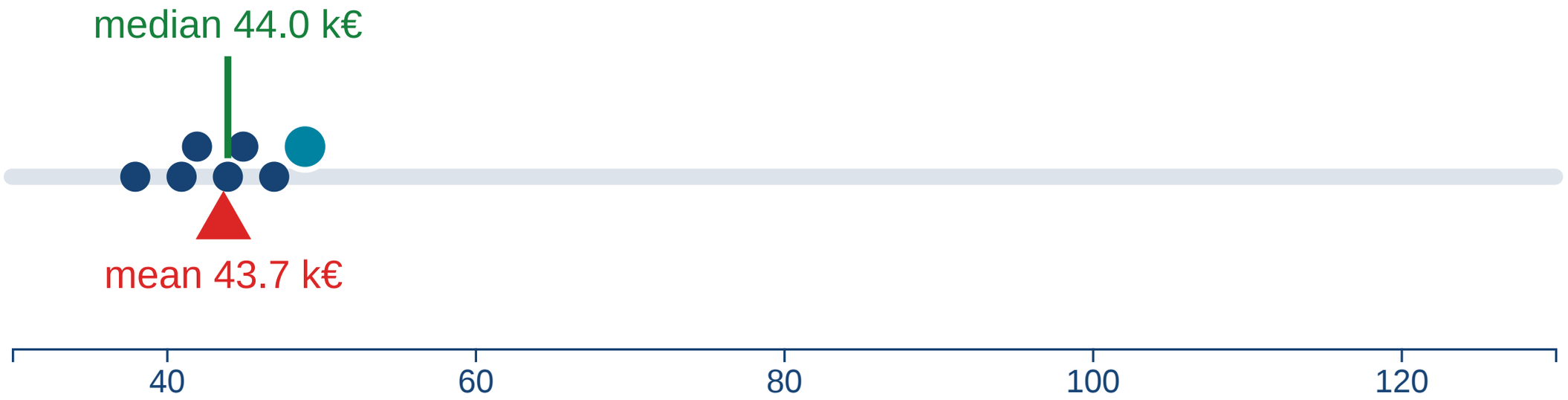
✓ **Robust** against outliers — one extreme value moves the median by at most one position

✗ **Costlier** to compute — it needs sorting or a selection algorithm, not one pass with a running sum

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Mini-Exercise — Solution

Mean ≈ 73.9, median $= 44$.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee; the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Sample variance and **standard deviation** of the observed data:

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2, \qquad s = \sqrt{s^2}$$

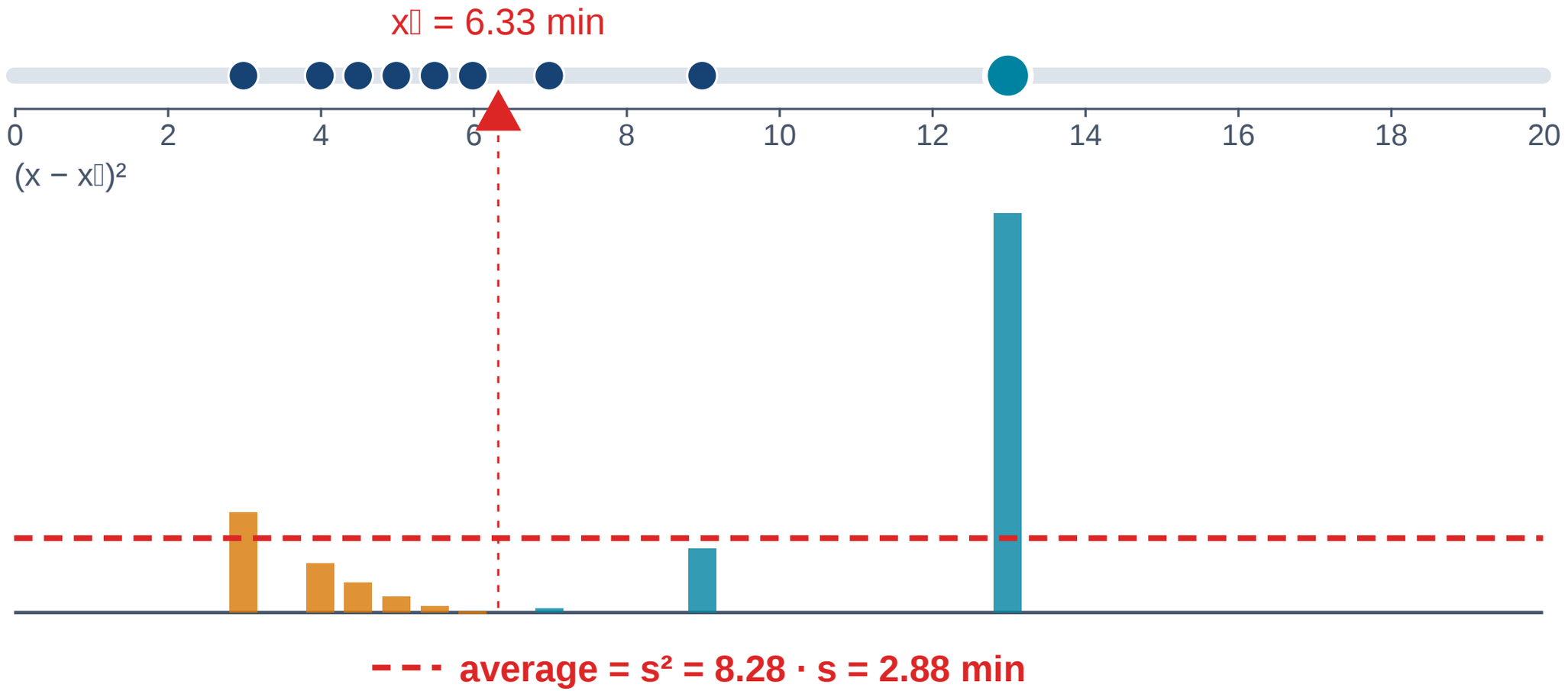
INTUITION

The average squared distance from the centre: how stretched or squeezed the data is. s brings it back to the unit of the data.

Computable in **one pass**, $O(n)$, from the running totals n, $\sum x_i$ and $\sum x_i^2$ — which matters when the data does not fit in memory.

The Variance, Drawn

The variance is the average square



Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. With the true mean μ unknown, it is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

INTUITION

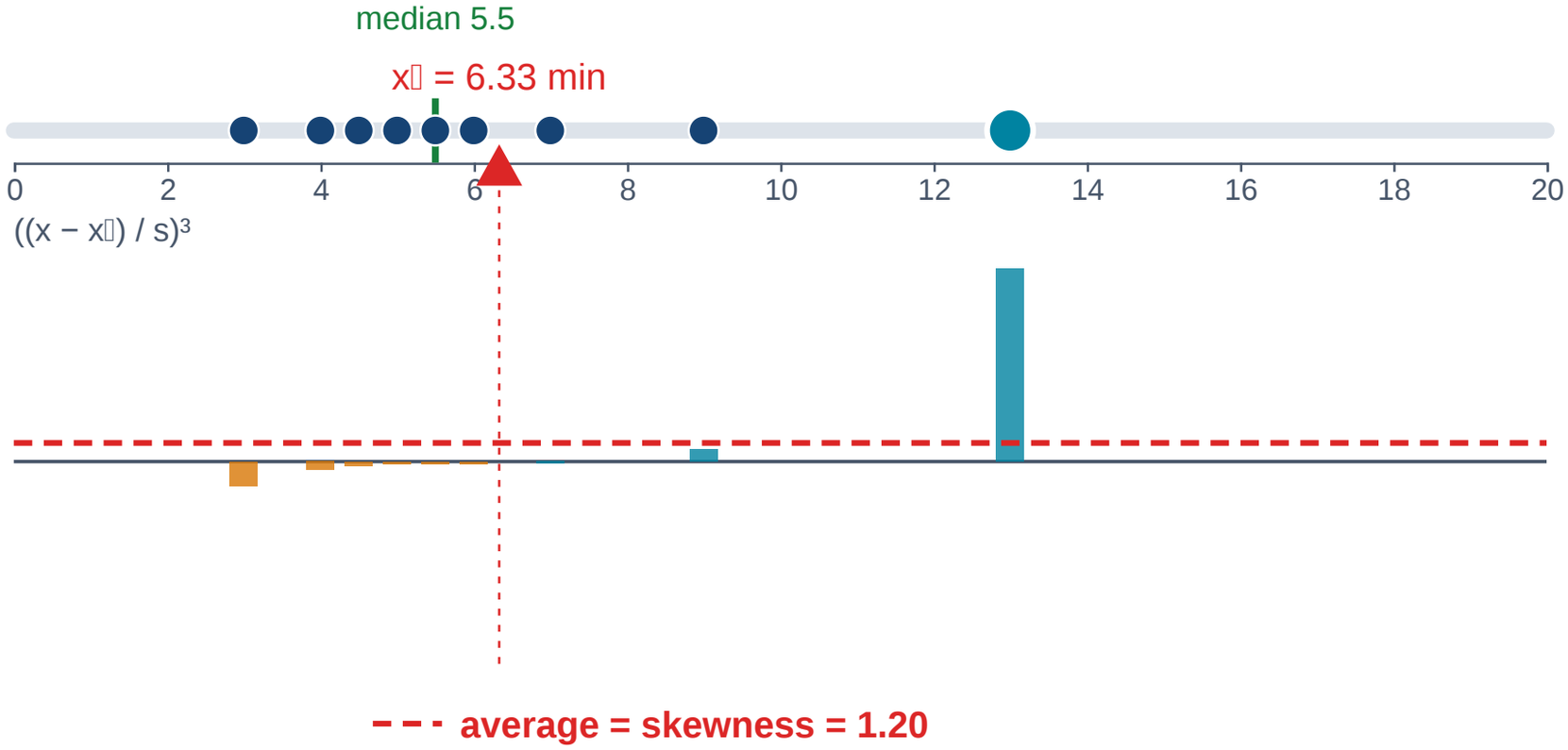
$\overline{x}$ is computed from the same data, so it sits closer to the data than μ does: the deviations come out **systematically too small**. Dividing by $n-1$ instead of n compensates — one degree of freedom was spent on the mean.

Naming is not uniform: many textbooks, and pandas, call the $n-1$ version s^2; NumPy's `var` divides by n unless told otherwise.

Sample Skewness: Which Tail Is Longer?

$$\widehat{\text{Skew}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \overline{x}}{s}\right)^3 \qquad \begin{cases} > 0 & \text{long right tail} \\ < 0 & \text{long left tail} \end{cases}$$

Right side outweighs left: skewed right



Drag the light point left. Where is the skewness 0 — and why there?

Takeaway

Variance and skewness average the deviations from the mean raised to the power **2 and 3**: squares add up, cubes compete by sign. s^2 describes **the sample**; $\hat{\sigma}^2$ estimates **the population** — the $n - 1$ is the price of having estimated the mean from the same data.

Quantiles

> The k-th q-**quantile** is the value x such that k/q of the values are smaller and $(q-k)/q$ are larger ($0 < k < q$).

Percentiles

$q = 100$

"the 12th percentile"

Quartiles

$q = 4$

Q_1 = 25th, Q_2 = 50th (= median),
Q_3 = 75th percentile

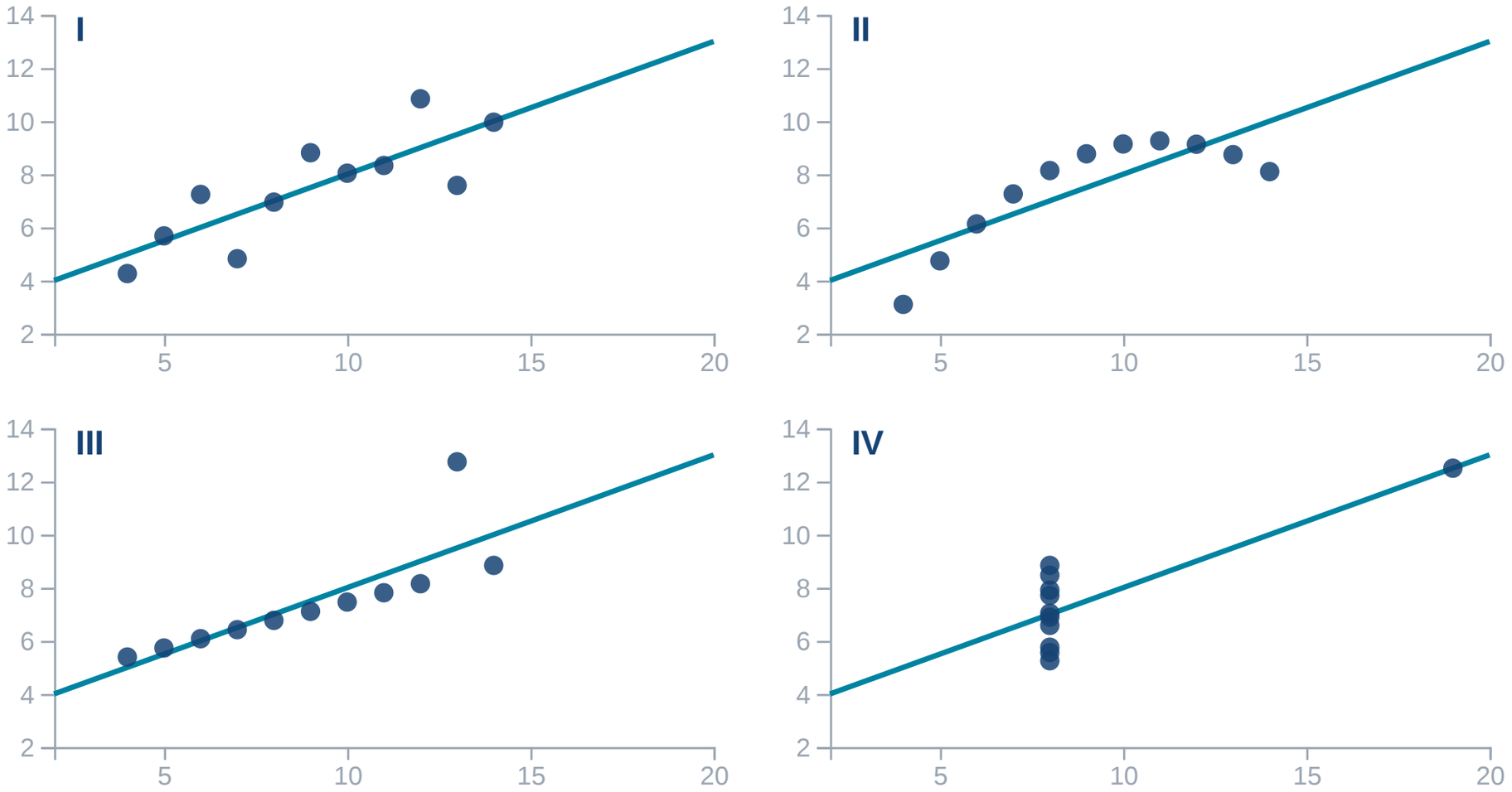
IQR

$$\text{IQR} = Q_3 - Q_1$$

the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data

all four: $\bar{x} = 9$, $s^2_x = 11$, $\bar{y} = 7.50$, $s^2_y \approx 4.12$, $r \approx 0.82$, $\hat{y} = 3.00 + 0.500 \cdot x$

What the Quartet Teaches

Summaries compress — and compression loses.
Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.
Visual inspection is not a soft alternative to statistics; it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

You Computed a Number. Which Number Did You Want?

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What is the probability that the population mean μ is exactly 73.9 k€?

Close to zero. $\overline{x}$ is not μ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is* μ but **how far from** μ **can this number be?**

Estimator and Estimate

Definition. An **estimator** is a function of the sample, $\hat{\theta} = g(X_1, \ldots, X_n)$, used to approximate an unknown quantity θ of the population. The number it takes on for one particular sample is an **estimate**.

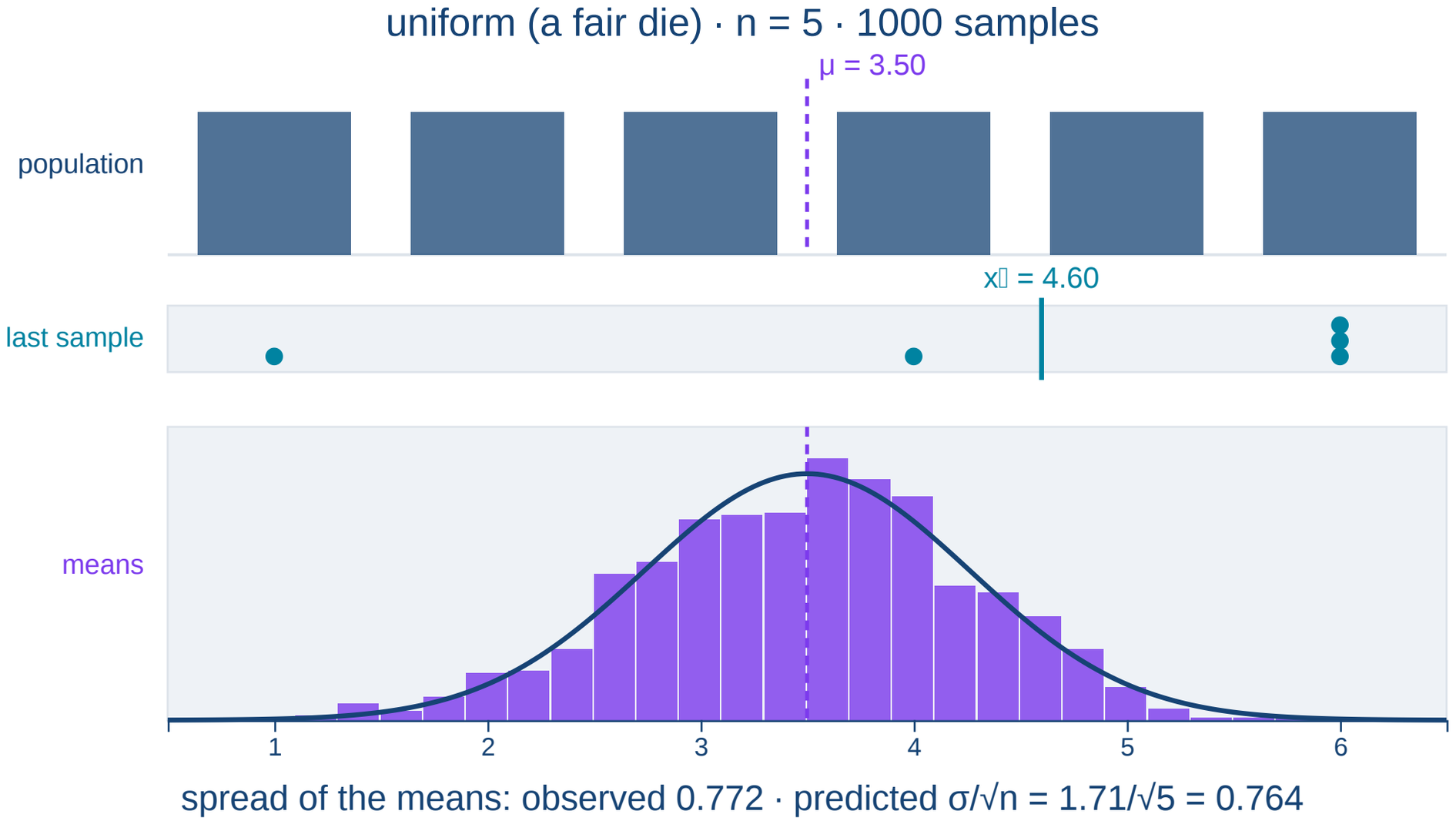
INTUITION

The estimator is the **recipe**, $\overline{X} = \frac{1}{n}\sum_i X_i$; the estimate is what one batch of data cooks from it, $\overline{x} = 73.9$.

The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

$\overline{x}$, s^2 and $\hat{\sigma}^2$ from descriptive statistics are all estimators — used so far without saying what kind of object they are.

An Estimator Has a Distribution



Three Things the Picture Shows

- → The sample means **centre on** μ — the estimator is not systematically off
- → They **scatter less than single draws**: standard deviation 0.76 instead of 1.71 — averaging cancels noise
- → Their histogram is **bell-shaped**, even though the population was not

> **Definition (Sampling distribution).** The distribution of an estimator $\hat{\theta}$ over all possible samples of size n. $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as n is more than a handful — which is why one formula can cover cases we never inspect.

Standard Error

Definition (Standard error). The standard error of an **estimator** $\hat{\theta}$ is the standard deviation of its **sampling distribution**: $\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$. For the sample mean of n independent draws, $\mathrm{SE}(\overline{X}) = \sigma/\sqrt{n}$.

Two distributions, two spreads — the die from the widget, $n = 5$:

	the population	the estimator $\overline{X}$
distribution of	one draw X	the mean of n draws, sample after sample
centre	$\mu = 3.5$	$\mu = 3.5$ (unbiased)
spread	$\sigma = 1.71$, the **standard deviation**	$\sigma/\sqrt{n} = 0.76$, the **standard error**
more data	unchanged	shrinks like $1/\sqrt{n}$

Precision Costs Data

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}} \quad \Longrightarrow \quad \frac{\sigma}{\sqrt{kn}} = \frac{1}{\sqrt{k}} \cdot \frac{\sigma}{\sqrt{n}}$$

Multiply the data by k, and the standard error is multiplied by $1/\sqrt{k}$:

0.71	→	½	→	⅕	→	1/10
2 × the data		4 × the data		25 × the data		100 × the data

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Two Theorems We Have Already Seen Happen

For i.i.d. observations with mean μ and finite variance σ^2:

Law of large numbers (LLN)	$\overline{X}_n \to \mu \text{ as } n \to \infty$	*the sample means close in on the truth*
Central limit theorem (CLT)	$\overline{X}_n \approx N(\mu, \sigma^2/n) \text{ for large } n$	*and we know the shape of what is left*

The first says the error goes away; the second says how much of it is left after n observations. Only the second lets us put a number on our ignorance.

From a Number to an Interval

Definition (Confidence interval). Statistics L, U of the sample form a **confidence interval** for θ at **level** $1-\alpha$ if, for every θ,

$$P(L \le \theta \le U) \ge 1-\alpha$$

Exact if $=$ holds for every θ.

Over **samples**: L, U random, θ fixed. Confidence is the **coverage rate** of the procedure, not a probability for θ.

z **from** α**:** $P(-z \le Z \le z) = 1-\alpha$ for $Z \sim N(0,1)$: 1.28, 1.96, 2.58 at 80, 95, 99%.

For the mean, the central limit theorem, $\overline{X} \approx N(\mu, \mathrm{SE}^2)$, gives $P(|\overline{X} - \mu| \le z\,\mathrm{SE}) \approx 1-\alpha$, hence

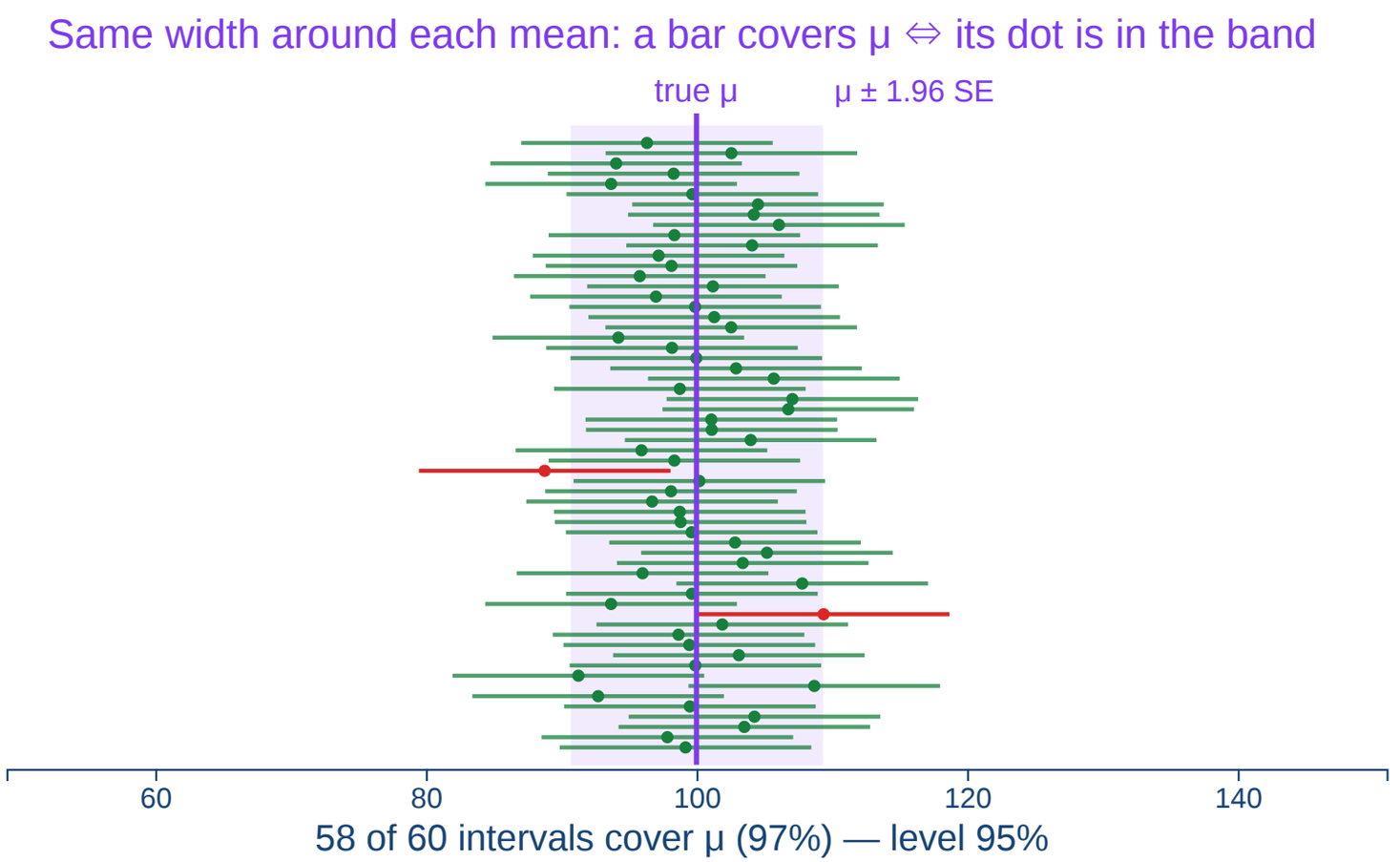
$$\hat{\theta} \pm z \cdot \mathrm{SE}(\hat{\theta})$$

the estimate, here x̄

the quantile set by α: 1.96 for 95%

the standard error, here σ/√n

What Does "95% Confident" Mean?



How to read it

- **Purple band:** $\mu \pm z \cdot \mathrm{SE}$ around the true $\mu = 100$; **confidence** sets z: 80 / 90 / 95 / 99 % → 1.28 / 1.64 / 1.96 / 2.58
- **Row:** one sample of n values (slider), $\sigma = 15$; **dot:** its mean $\overline{x}$, not a data point — 10, then 60 samples
- **Green / red:** $\overline{x}$ inside / outside the band
- **Bar** (last step): $\overline{x} \pm z \cdot \mathrm{SE}$ — it covers μ exactly when its dot is inside the band
- Higher confidence → wider band and bars → fewer misses

Confidence Is Coverage, Not Certainty

One sample, estimating $\theta = \mu$ with $\hat{\theta} = \overline{x} = 103$, $\mathrm{SE}(\hat{\theta}) = 3$:

Which of the three tells you the most about θ?

level	$\hat{\theta} \pm z \cdot \mathrm{SE}(\hat{\theta})$	width
80%	$[99.2,\ 106.8]$	7.7
95%	$[97.1,\ 108.9]$	11.8
99%	$[95.3,\ 110.7]$	15.5

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80%	$[99.2,\ 106.8]$	7.7
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99%	$[95.3,\ 110.7]$	15.5

Which of the three tells you the most about θ? None — the data are the same. The level is a **promise**: the rule may miss in a share α of samples. A **weaker promise** allows a **narrower interval**.

- **100%:** $(-\infty, \infty)$ — always right, says nothing
- **near 0%:** the point $\hat{\theta}$ — precise, almost never right

At the same level, only **more data** narrows it: $\mathrm{SE}(\hat{\theta}) = \sigma/\sqrt{n}$.

The Sentence Almost Everyone Gets Wrong

WHAT IT DOES NOT MEAN

"There is a 95% probability that θ lies in **this** interval."

θ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

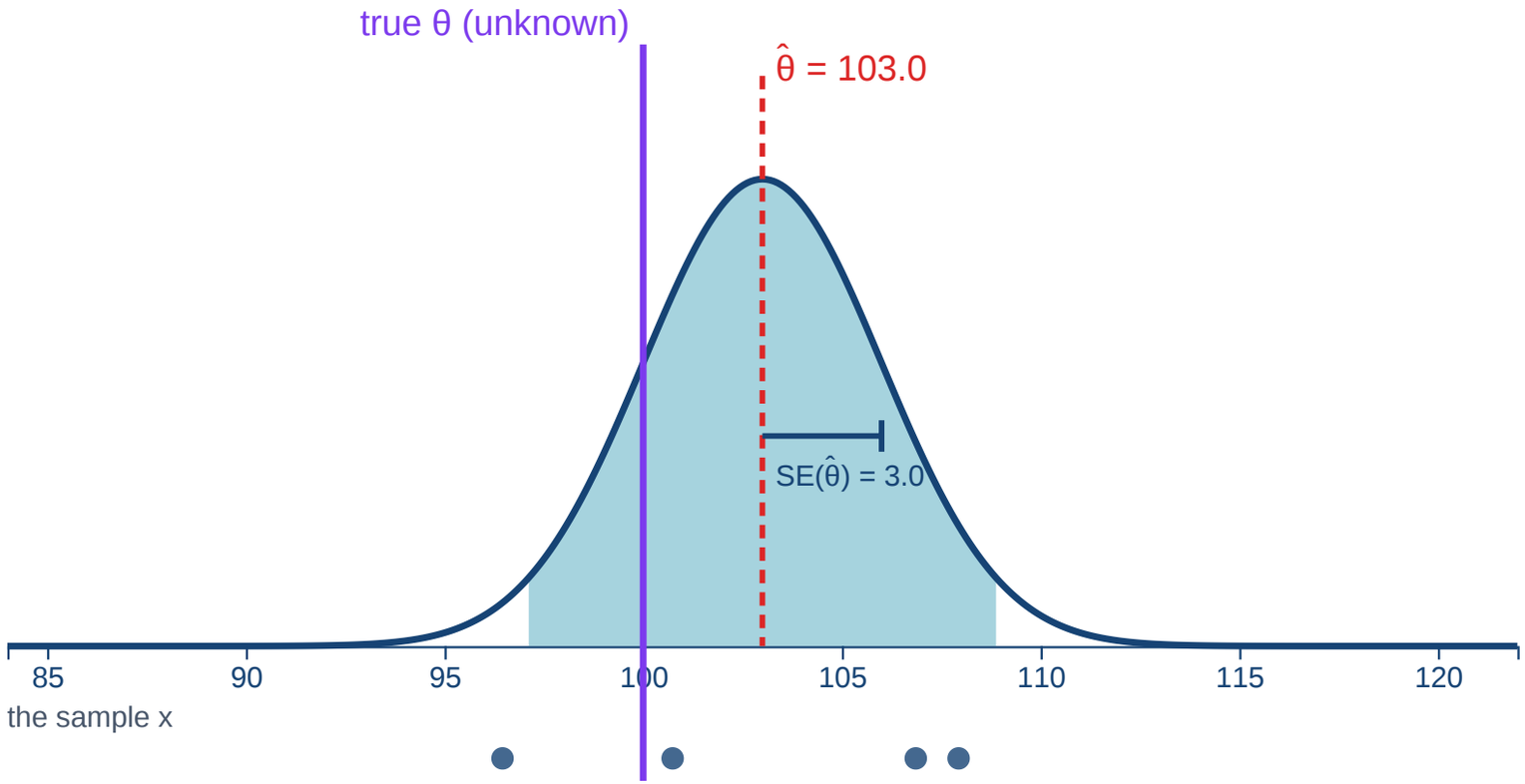
WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain θ."

The probability is a property of the recipe, evaluated before the data is seen.

The Bayesian Reading: A Distribution Over θ

Draw a new sample, or add data: what moves, and what does not?



n = 4 · θ̂ = 103.0 · SE(θ̂) = 3.00 · 95% credible [97.1, 108.9] · contains θ

Bayes, for θ and the sample $\mathbf{x} = (x_1, \ldots, x_n)$:

$$p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta)\, p(\theta)$$

With a flat prior: $\theta \mid \mathbf{x} \sim N(103, 3^2)$.

> "95% probability that θ lies in $[97.1, 108.9]$": false for the confidence interval, **correct** for the credible interval, $P(97.1 \le \theta \le 108.9 \mid \mathbf{x}) = 0.95$. Also $P(\theta > 100 \mid \mathbf{x}) \approx 0.84$.

Frequentist or Bayesian: What Is Uncertain?

	frequentist	Bayesian
the parameter θ	fixed, unknown — **no** probability	fixed, unknown — uncertainty about it as a **distribution** $p(\theta \mid \mathbf{x})$
the sample $\mathbf{x}$	random before it is seen, and so is $\hat{\theta}$	observed, conditioned on
the interval	confidence interval $\hat{\theta} \pm z \cdot \mathrm{SE}(\hat{\theta})$	credible interval
"95%" means	95% of intervals from repeated samples contain θ	$P(\theta \in \text{this interval} \mid \mathbf{x}) = 0.95$
needs	a sampling model	a sampling model **and** a prior $p(\theta)$

> Both still trade coverage for width: a 99% credible interval is wider than an 80% one. Neither gets more information out of the same sample.

Where This Leaves Us

Two kinds of uncertainty meet in one formula. We can now measure one of them.

- → σ — how variable the world is. **Aleatoric.** Unchanged by data.
- → $\mathrm{SE} = \sigma/\sqrt{n}$ — how badly we know our own estimate. **Epistemic.** Shrinks with data.

"More data helps" was a slogan.
It is now a rate: $1/\sqrt{n}$.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

$\mathrm{SE} = 60/\sqrt{100} = 6$ s, so the 95% interval is about 180 ± 12 s. Halving the SE needs twice the $\sqrt{n}$ — **four times the data**, $n = 400$. $\sigma = 60$ s does not change: customers stay as variable as they were; only our knowledge of their average improves.

Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you. A probability for θ itself needs the Bayesian reading: a prior, and a **credible interval** from the posterior.

Module 7

Hypothesis Testing

> ***Guiding Question:*** *Your new model scores higher than the baseline. How do you know the improvement is real?*

Motivation: Comparing Two ML Experiments

Experiment A — baseline model accuracy measured over n_A runs

Experiment B — new architecture accuracy measured over n_B runs

B scored higher. Is the improvement systematic, or just randomness?

H_0: the two means are equal — $\mu_A = \mu_B$
H_1: they differ — $\mu_A \neq \mu_B$

The Idea

> Compare what we **observed** with what H_0 **predicts**. If the disagreement is large enough to be implausible under H_0, reject H_0.

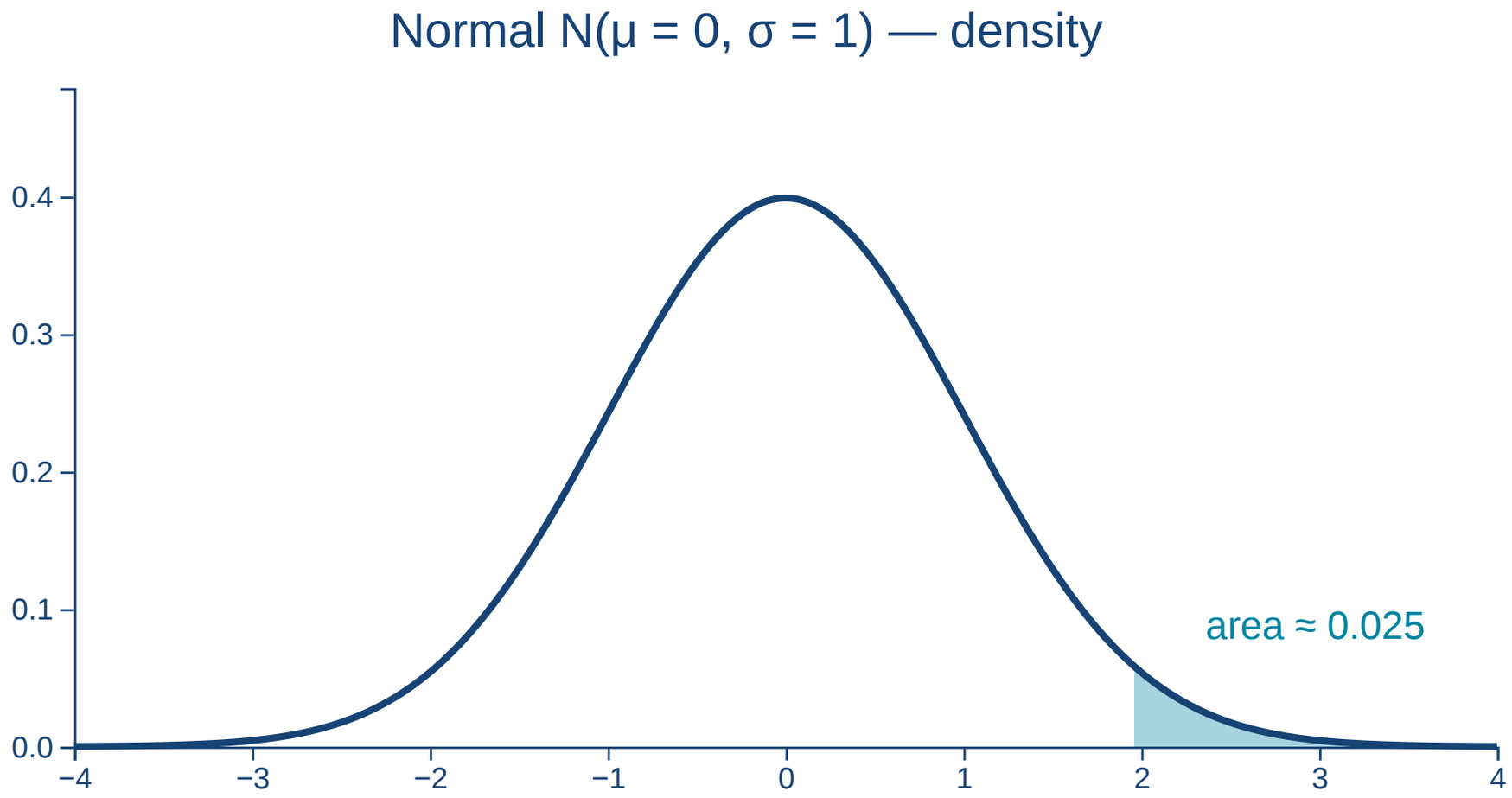
We never prove H_1. We only measure how badly H_0 fails to explain the data.

Test Statistic and p-Value

A **test statistic** $T = T(X_1, \ldots, X_n)$ compresses the data so that its distribution **under** H_0 is known.
The **p-value** is the probability, under H_0, of observing a statistic at least as extreme as the one we got.
Small p-value → the data is surprising under H_0.

> The p-value is **not** the probability that H_0 is true. It is $P(|T| \geq |t_{\text{obs}}| \mid H_0)$ — a tail probability of the statistic — not $P(H_0 \mid \text{data})$.

The Rejection Region



Under H_0 the statistic follows a known distribution. The shaded tail carries probability $\alpha/2$: if T lands there, we reject.

Two Ways to Be Wrong

Type I error — false alarm
Reject H_0 when it is true.
Probability α (we choose it).

Type II error — miss
Fail to reject H_0 when H_1 holds.
Probability β; **power** $= 1 - \beta$.

Lowering α raises β for a fixed design. To improve both: more data, less noise, or a **paired** design that cancels shared noise.

The same rule as the confidence interval: reject $H_0 : \theta = \theta_0$ at level $\alpha \iff \theta_0$ lies outside the $1 - \alpha$ interval. Type I = the interval misses the true θ; Type II = it contains a wrong θ_0.

One Sample: z or t?

Known variance — z-test

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$$

Reject (two-sided) if $|Z| > z_{1-\alpha/2}$.

Unknown variance — t-test

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$

s estimated from the data → **heavier tails** (t with $n-1$ d.f.).

$\sigma/\sqrt{n}$ is the **standard error**: the standard deviation of the sample mean $\overline{X}$. $s = \sqrt{s^2}$ is the sample standard deviation. Both statistics measure the distance from μ_0 in standard errors.

Two Samples: Which t?

Situation	Test	Statistic
independent runs, equal variances credible	pooled t	$\frac{\overline{X}_B - \overline{X}_A}{s_p\sqrt{1/n_A + 1/n_B}}$
independent runs, variances may differ	**Welch's t**	$\frac{\overline{X}_B - \overline{X}_A}{\sqrt{s_A^2/n_A + s_B^2/n_B}}$
both models on the **same** seeds or splits	**paired t**	t-test on the differences $d_i = x_{B,i} - x_{A,i}$

> Welch is the default for independent runs. When both models can be run on the same splits, pair them: the shared noise cancels in d_i.

Testing Variances

Single variance (normal model), χ^2 statistic with $n-1$ degrees of freedom.

Is run-to-run variability within specification?

Two variances, $F = s_A^2 / s_B^2$.

If the variances differ, **do not pool —** go back to Welch's t for the means.

Back to the A/B Question — Worked

Ten runs each. A: mean accuracy 80 %, $s_A = 2$. B: mean 82 %, $s_B = 1$.

$$t = \frac{82 - 80}{\sqrt{4/10 + 1/10}} = \frac{2}{0.71} \approx 2.83$$

Back to the A/B Question — Worked

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Welch's degrees of freedom ≈ 13; the critical value $t_{0.975} \approx 2.16$. Since $2.83 > 2.16$, **reject** H_0 at $\alpha = 0.05$ ($p \approx 0.014$).

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> Paired on the same ten seeds, with differences of mean 2 and $s_d = 0.5$: $t = 2/(0.5/\sqrt{10}) \approx 12.6$. Pairing buys far more power than more runs.

The Recommendation

Default

Welch's two-sample t on the mean difference — or paired t if both ran on the same splits

Care about stability

F-test on s_A^2 / s_B^2

Normality dubious

Permutation (randomisation) test on the mean difference

Hypothesis testing compares data to a model through a statistic whose null distribution is known. Known vs. unknown variance changes that distribution (z vs. t); the design changes the test (Welch for independent runs, paired for shared splits).

The Churn Shop: Is the Difference Real?

Lecture 01's dashboard, with counts. Over one year:

	customers	churned	rate
app users	400	40	10%
no app	400	60	15%

H_0: both groups churn at the same rate. Pooled rate $\hat{p} = 100/800 = 0.125$.

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$$z = \frac{0.15 - 0.10}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{400} + \frac{1}{400}\right)}} = \frac{0.05}{0.023} \approx 2.1 \qquad p \approx 0.03$$

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> Chance alone would rarely produce a gap this large: at $\alpha = 0.05$ the difference is **real**. Whether the app *causes* it is a different question — Lecture 03 draws both answers.

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.